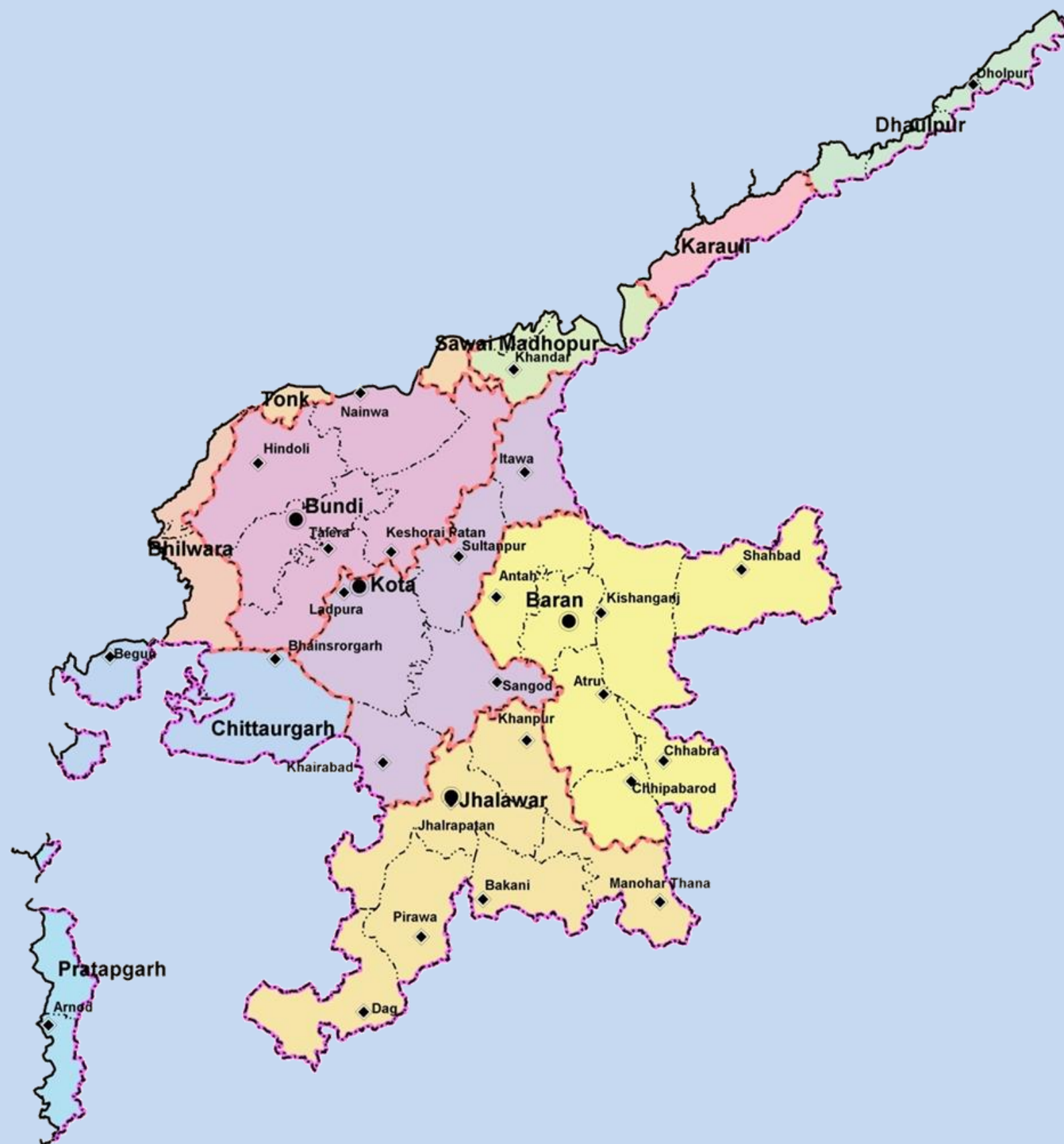


# Hydrogeological Atlas of Rajasthan

## Chambal River Basin



2013

## Message



**Shri Ashok Gehlot,**  
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

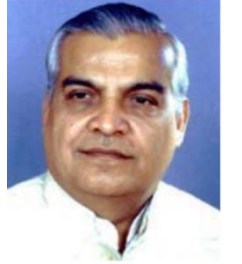
I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of 'Hydrogeological Atlas of Rajasthan'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

**(Ashok Gehlot)**

## Message



**Dr. Jitendra Singh,**  
Hon'ble Minister for Energy,  
Non-Conventional Energy Resources,  
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

**(Jitendra Singh)**

## Message



**Dr. Purushottam Agarwal**

Principal Secretary to Government,  
Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of '*Hydrogeological Atlas of Rajasthan*' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

## Foreword



**Arno Schaefer**

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

**(Arno Schaefer)**

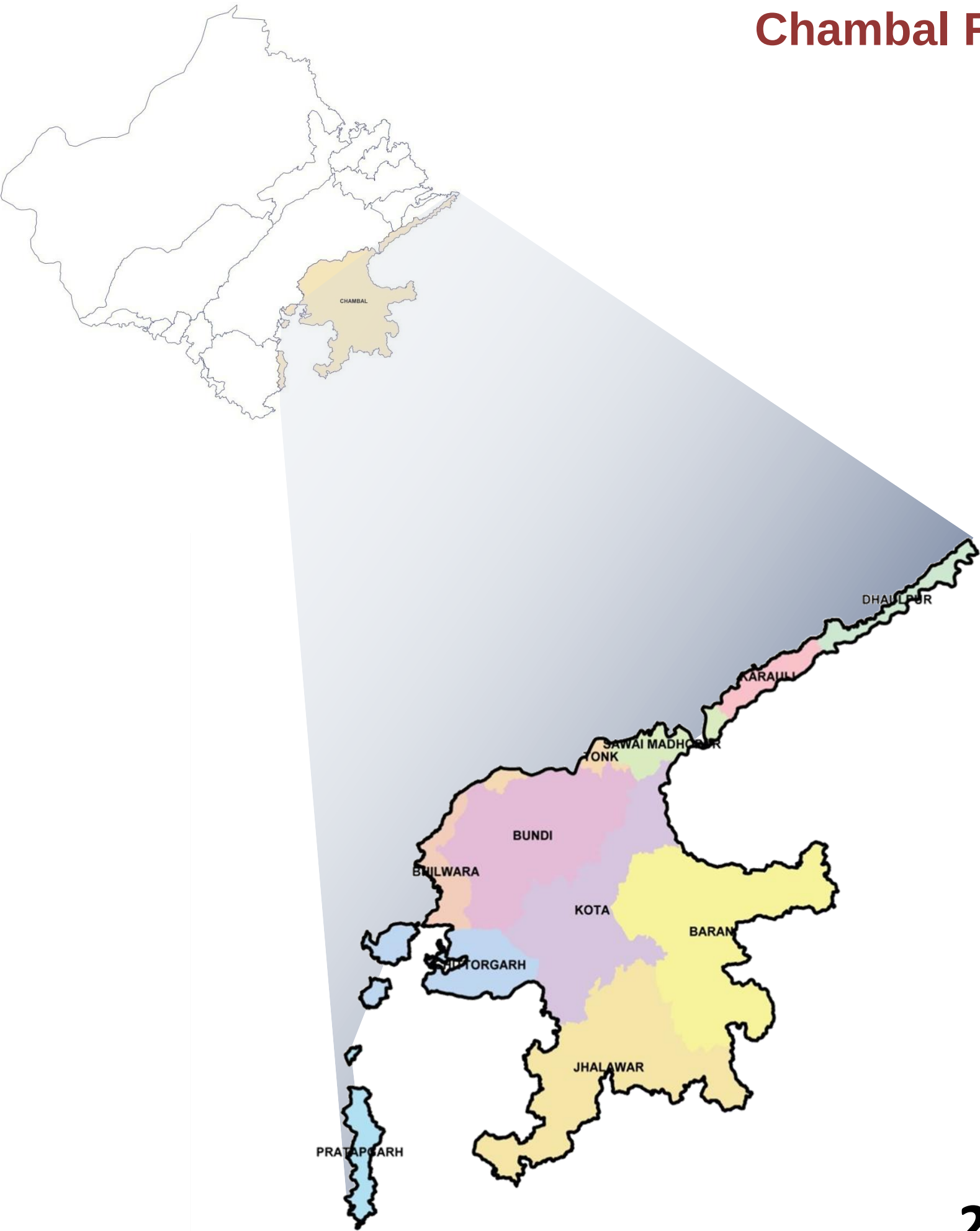


# Hydrogeological Atlas of Rajasthan

## Chambal River Basin

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## ADMINISTRATIVE SETUP

## CHAMBAL RIVER BASIN

### Location:

Chambal River Basin is located in the south-eastern part of Rajasthan. It stretches between 23° 31' 03.91" to 26° 56' 03.72" North latitudes and between 74° 45' 15.00" to 78° 13' 33.86" East longitudes. It is bounded primarily by the Banas river basin in the west, partly by Mahi, Gambhir and Parbati river Basins. The eastern boundary is defined by Madhya Pradesh State where the rest of the catchment area of Chambal river lies. The Chambal basin extends over parts of Baran, Bhilwara, Bundi, Chittaurgarh, Dhaulpur, Jhalawar, Karauli, Kota, Pratapgarh, Sawai Madhopur and Tonk Districts. The total catchment area of the basin is approximately 31,239 km<sup>2</sup> within the state of Rajasthan (excluding Banas river basin which is tributary to Chambal River). River Chambal is a principal tributary of river Yamuna and originates in the Vindhyan ranges near Mhow in Indore District of Madhya Pradesh, entering through a deep gorge in Rajasthan at Chourasigarh, about 96 km upstream of Kota. The deep gorge extends up to Kota and the river then flows for about 226 km within Rajasthan in a north-easterly direction, and then forms the boundary between MP and Rajasthan for about 252 km.

### Administrative Set-up:

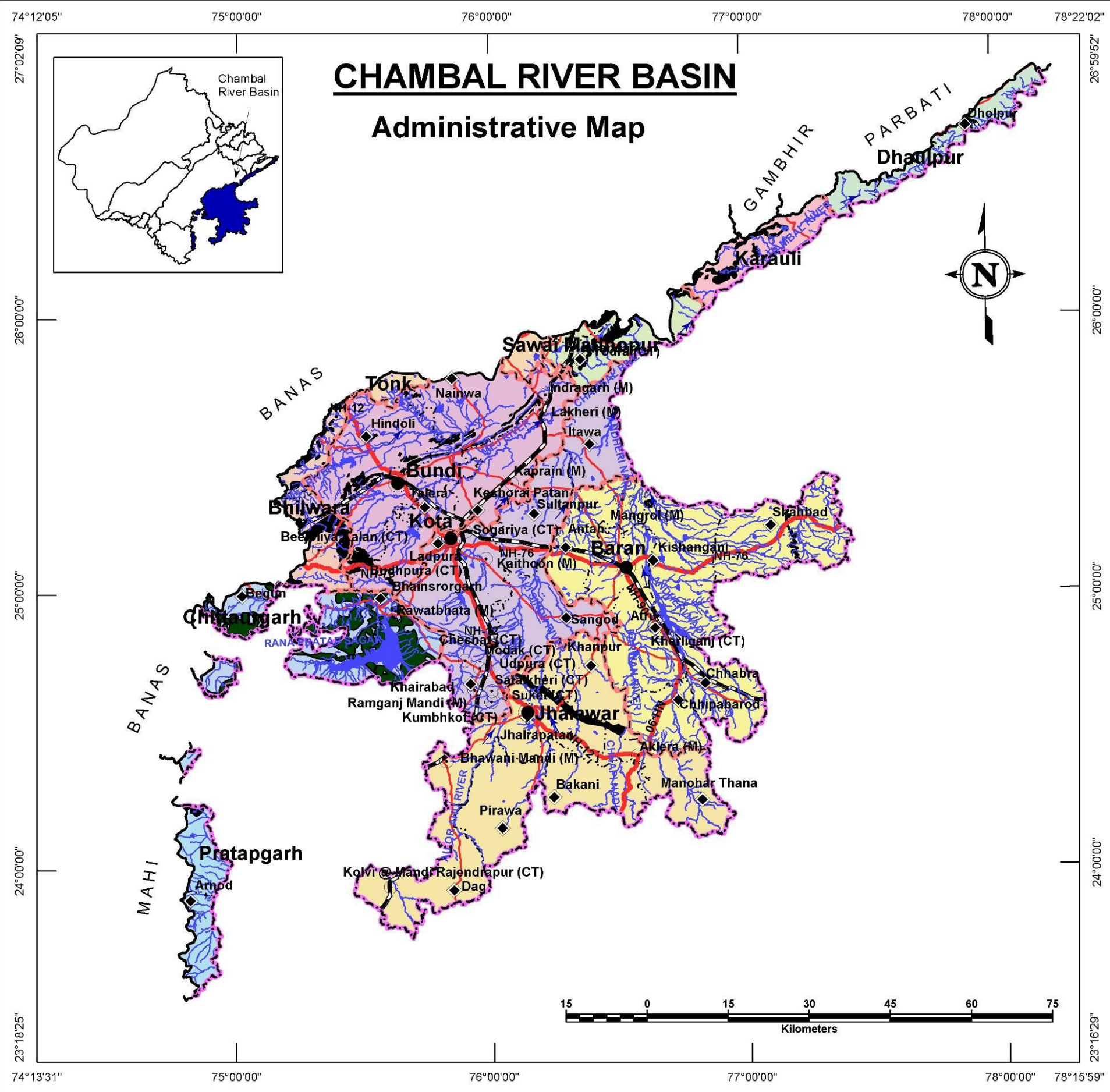
Administratively, Chambal river basin extends over parts of Baran, Bhilwara, Bundi, Chittaurgarh, Dhaulpur, Jhalawar, Karauli, Kota, Pratapgarh, Sawai Madhopur and Tonk districts encompassing 41 Blocks and 5,786 towns and villages.

| S. No.       | District Name  | Area (sq km)    | % of Basin Area | Total Number of Blocks | Total Number of Towns and Villages |
|--------------|----------------|-----------------|-----------------|------------------------|------------------------------------|
| 1            | Baran          | 6,993.9         | 22.4            | 7                      | 1,213                              |
| 2            | Bhilwara       | 1,045.3         | 3.4             | 2                      | 219                                |
| 3            | Bundi          | 5,610.2         | 18.0            | 5                      | 827                                |
| 4            | Chittaurgarh   | 2,197.9         | 7.0             | 3                      | 359                                |
| 5            | Dhaulpur       | 844.9           | 2.6             | 4                      | 151                                |
| 6            | Jhalawar       | 6,315.3         | 20.2            | 6                      | 1,608                              |
| 7            | Karauli        | 816.0           | 2.6             | 2                      | 82                                 |
| 8            | Kota           | 5,122.3         | 16.4            | 5                      | 903                                |
| 9            | Pratapgarh     | 895.2           | 2.9             | 3                      | 218                                |
| 10           | Sawai Madhopur | 952.2           | 3.1             | 2                      | 121                                |
| 11           | Tonk           | 446.0           | 1.4             | 2                      | 85                                 |
| <b>Total</b> |                | <b>31,239.2</b> | <b>100.0</b>    | <b>41</b>              | <b>5,786</b>                       |

### Climate:

Chambal river basin falls within Semi-arid to Sub-humid climatic regions. It is very cold from November to February while turning hot from March to June when the maximum temperature rises up to 48°C. The mean annual rainfall over basin is approximately 697 mm (year 2010). Most of the rainfall is received during the Monsoon months between June/July to September. Months of September and October are very pleasant before onset of winters.

PLATE - I



## LEGEND

### Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

### River Basin Boundary



### Roads:

- National Highway
- State Highway

### Railway:

- Broad Gauge
- Metre Gauge

### Water Bodies:

- River / Streams
- Ponds / Reservoirs

### Hills



### Reserve Forest



## TOPOGRAPHY

High elevation areas surround the central low lying parts of basin from east, south and western sides. The river flows in the north eastern direction and a narrow part extends through Karauli and Dhaulpur districts. The minimum elevation in the basin is 111 m amsl in Dhaulpur district where the river drains out of the basin and the maximum elevation is approximately 605 m amsl in Bhilwara district. Nearly half of the basin is hilly while rest is undulating plains.

**Table: District wise minimum and maximum elevation**

| District Name  | Min. Elevation (m amsl) | Max. Elevation (m amsl) |
|----------------|-------------------------|-------------------------|
| Baran          | 196.0                   | 547.3                   |
| Bhilwara       | 304.3                   | 605.3                   |
| Bundi          | 186.6                   | 547.0                   |
| Chittaurgarh   | 293.6                   | 604.1                   |
| Dhaulpur       | 111.3                   | 341.2                   |
| Jhalawar       | 245.9                   | 523.9                   |
| Karauli        | 131.0                   | 469.5                   |
| Kota           | 177.0                   | 517.0                   |
| Pratapgarh     | 437.0                   | 575.0                   |
| Sawai Madhopur | 164.7                   | 502.7                   |
| Tonk           | 217.1                   | 473.4                   |

## RAINFALL

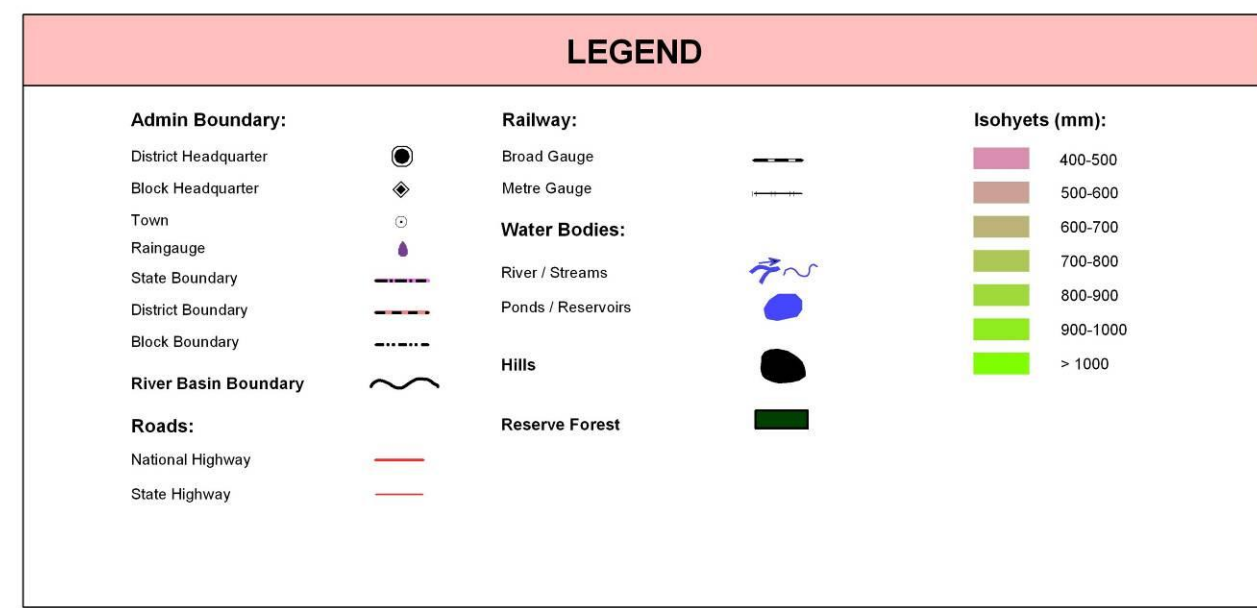
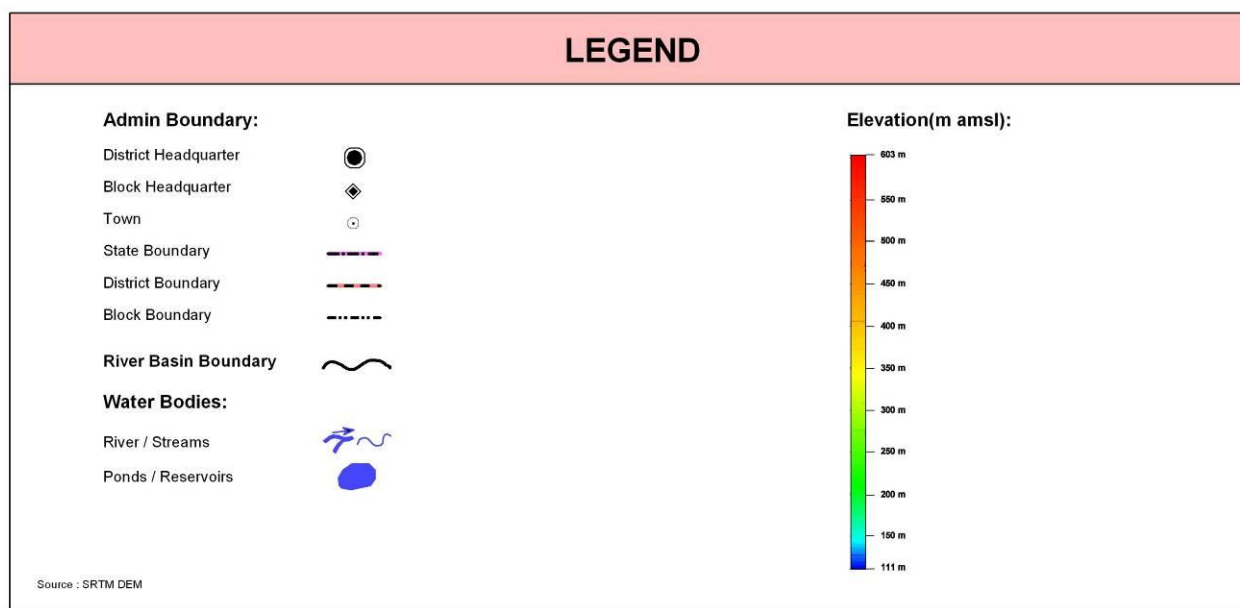
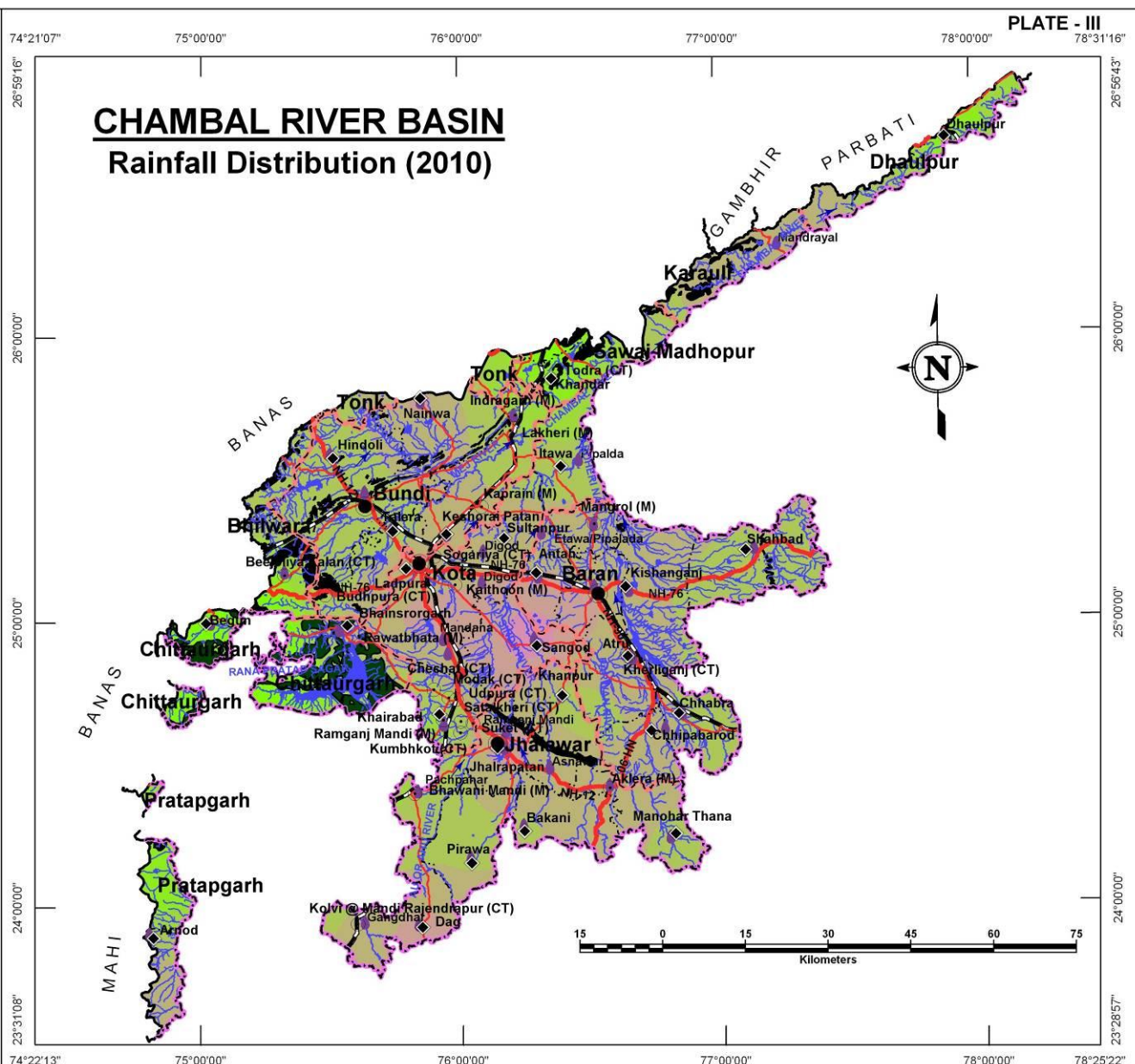
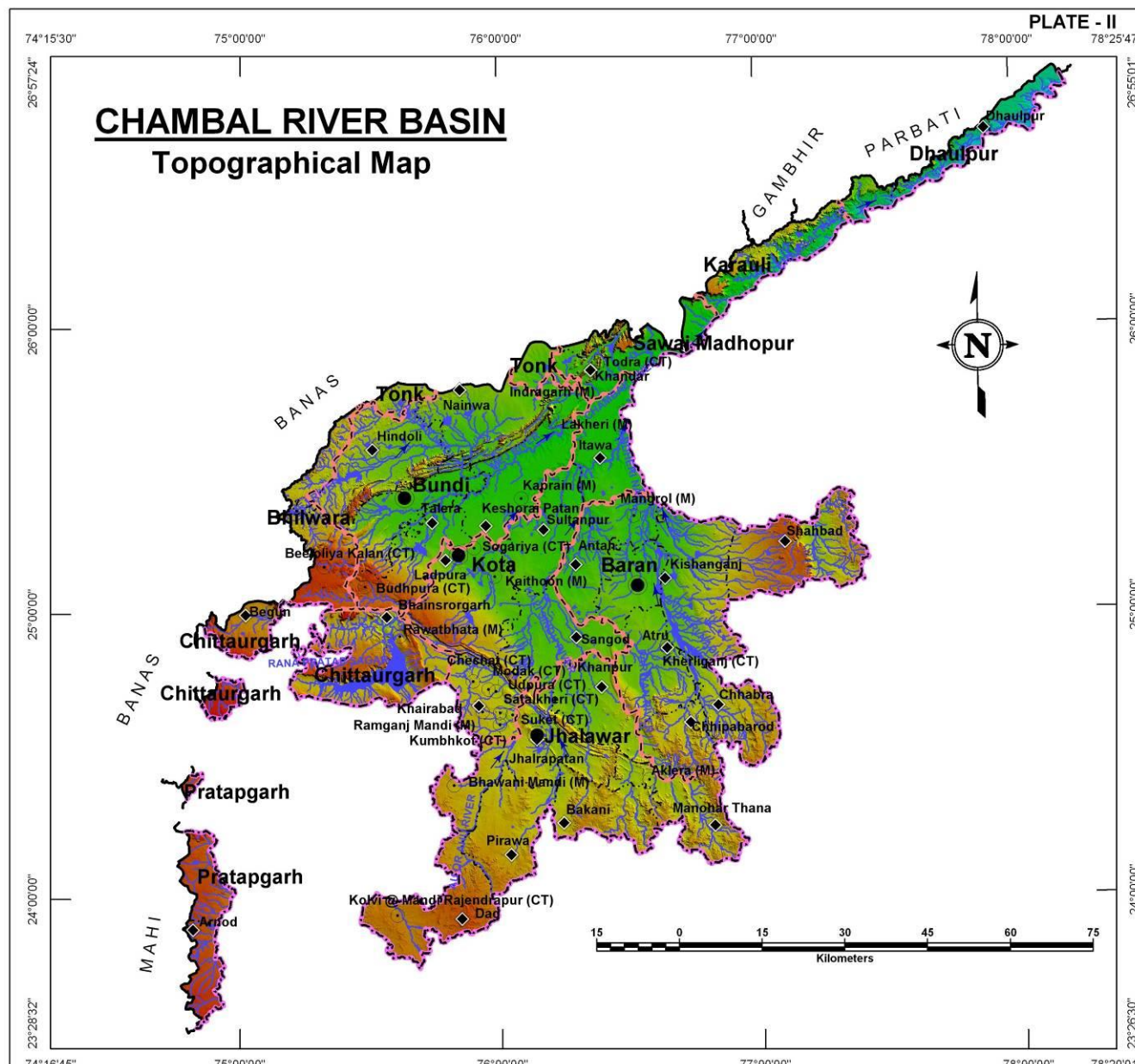
The general distribution of rainfall across the Chambal river basin can be visualized from isohyets presented in the Plate III where most of the hilly areas seem to receive higher rainfalls (total annual rainfall in the range of 700-900mm) whereas, the central part received lower rainfall (between 400 to 700mm). The average annual rainfall for the year 2010, computed based on available station data is about 697 mm and Begun block had received highest rainfall measuring about 1,790 mm followed by Bijolia, 880 mm. The rainfall data for available rain gauge stations is presented below. Minimum rainfall was recorded in Sangod block (415.1 mm).

**Table: District wise total annual rainfall (based on year 2010 meteorological station recordings (<http://waterresources.rajasthan.gov.in>))**

| S. No. | Rain gauge Stations | Total Monsoon Rainfall (mm) | Total Non-Monsoon Rainfall (mm) | Total Annual Rainfall (mm) |
|--------|---------------------|-----------------------------|---------------------------------|----------------------------|
| 1      | Antah               | 498.0                       | 85.0                            | 583.0                      |
| 2      | Atru                | 615.0                       | 53.0                            | 668.0                      |
| 3      | Bakani              | 660.5                       | 60.8                            | 721.3                      |
| 4      | Baran               | 560.0                       | 54.0                            | 614.0                      |
| 5      | Begun               | 1,624.0                     | 164.0                           | 1,788.0                    |
| 6      | Bhainsrodgarh       | 623.0                       | 70.0                            | 693.0                      |
| 7      | Bijolia             | 787.0                       | 93.0                            | 880.0                      |
| 8      | Chhabra             | 529.0                       | 33.0                            | 562.0                      |
| 9      | Chhipabarod         | 650.0                       | 38.0                            | 688.0                      |
| 10     | Digod               | 554.5                       | 84.0                            | 638.5                      |
| 11     | Dug                 | 518.0                       | 41.0                            | 559.0                      |
| 12     | Hindoli             | 658.0                       | 118.0                           | 776.0                      |

| S. No. | Rain gauge Stations | Total Monsoon Rainfall (mm) | Total Non-Monsoon Rainfall (mm) | Total Annual Rainfall (mm) |
|--------|---------------------|-----------------------------|---------------------------------|----------------------------|
| 13     | Jhalrapatan         | 484.6                       | 40.0                            | 524.6                      |
| 14     | Keshoraipatan       | 591.0                       | 87.0                            | 678.0                      |
| 15     | Khanpur             | 679.2                       | 67.0                            | 746.2                      |
| 16     | Kishanganj          | 549.0                       | 53.0                            | 602.0                      |
| 17     | Ladpura             | 515.0                       | 56.0                            | 571.0                      |
| 18     | Manoharthana        | 514.1                       | 62.6                            | 576.7                      |
| 19     | Nainwa              | 489.0                       | 79.0                            | 568.0                      |
| 20     | Pirawa              | 665.0                       | 54.0                            | 719.0                      |
| 21     | Ramganjmandi        | 828.0                       | 19.0                            | 847.0                      |
| 22     | Sangod              | 356.1                       | 59.0                            | 415.1                      |
| 23     | Shahbad             | 535.0                       | 39.5                            | 574.5                      |
| 24     | Talera              | 595.0                       | 137.0                           | 732.0                      |





## GEOLOGY

## CHAMBAL RIVER BASIN

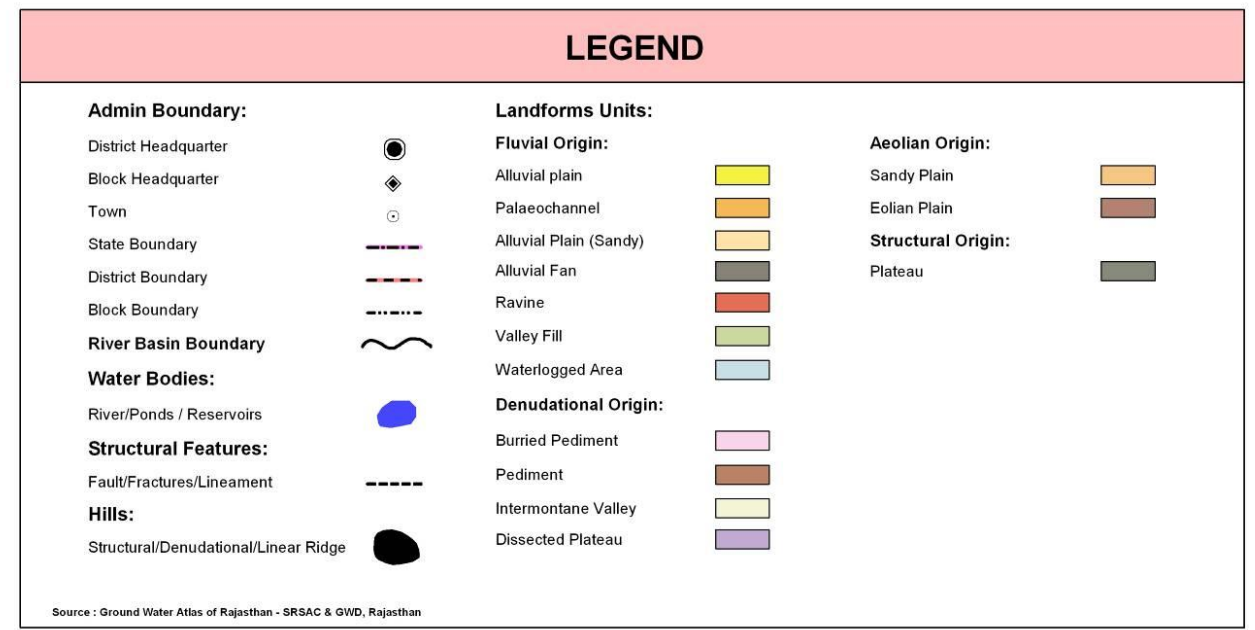
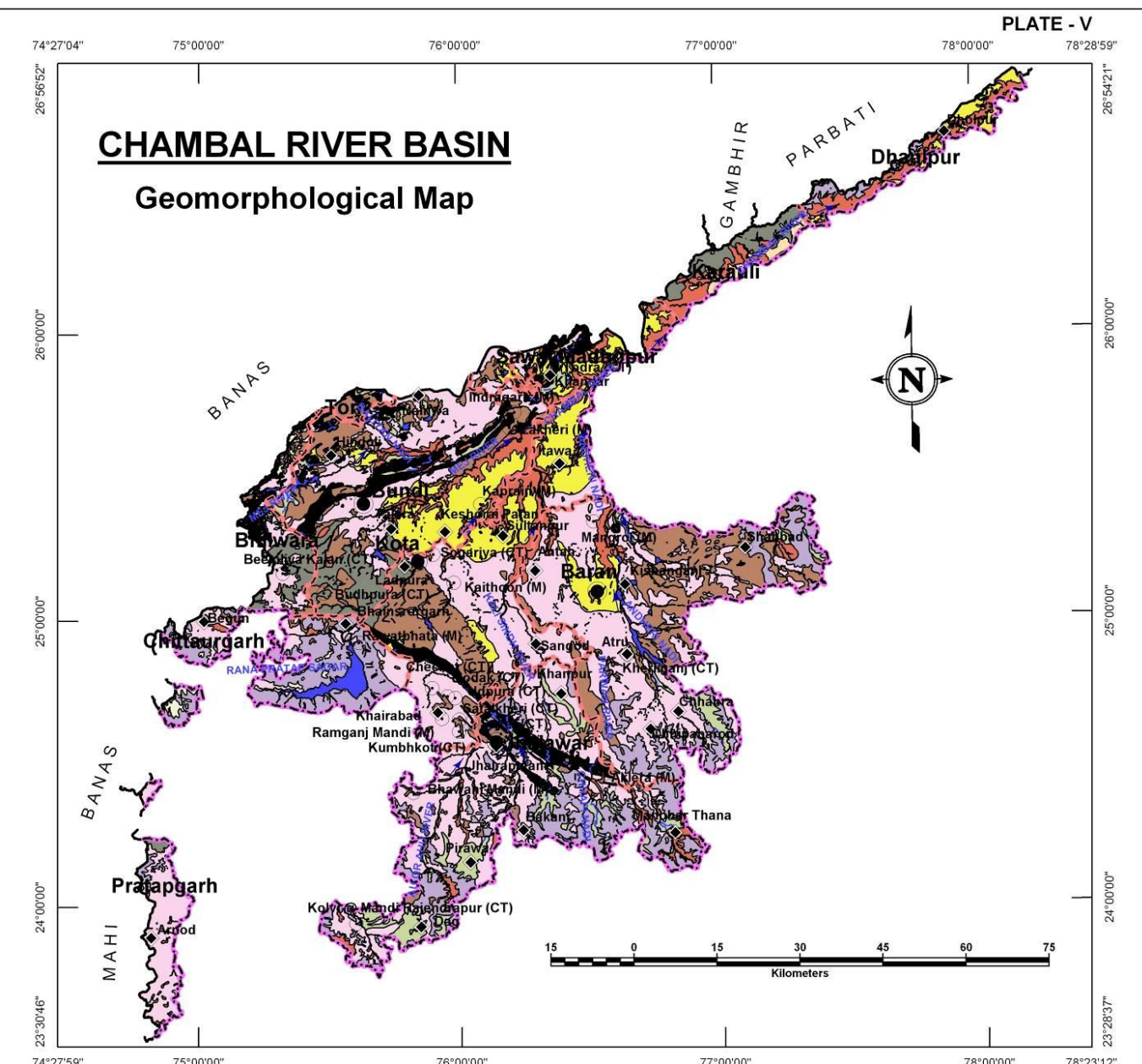
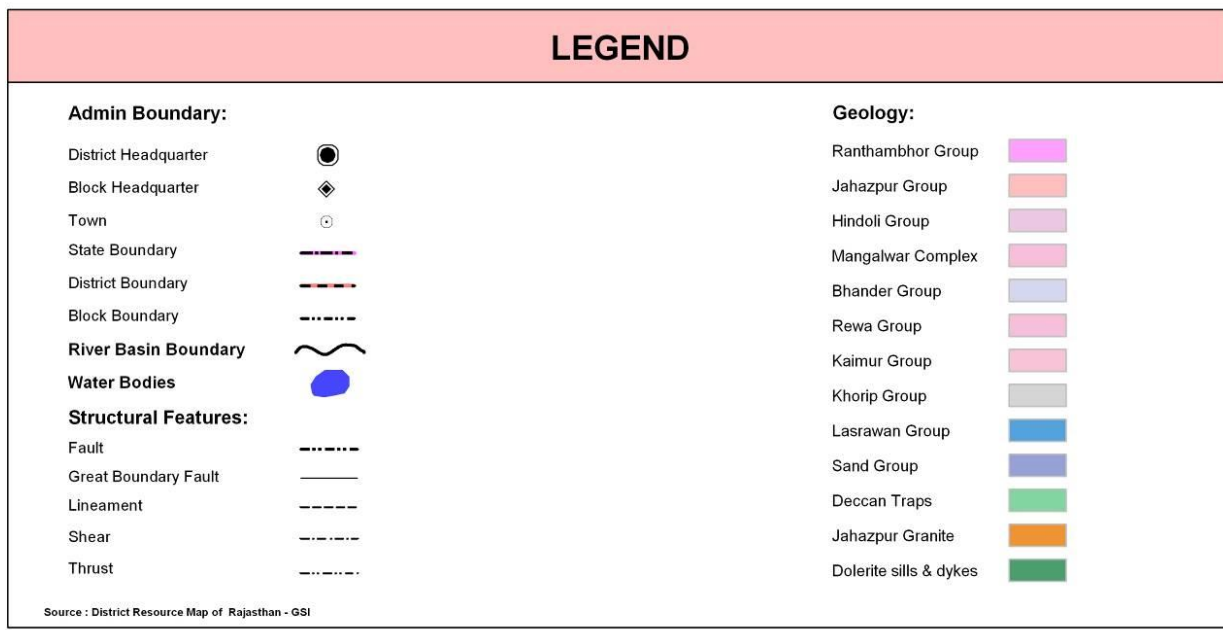
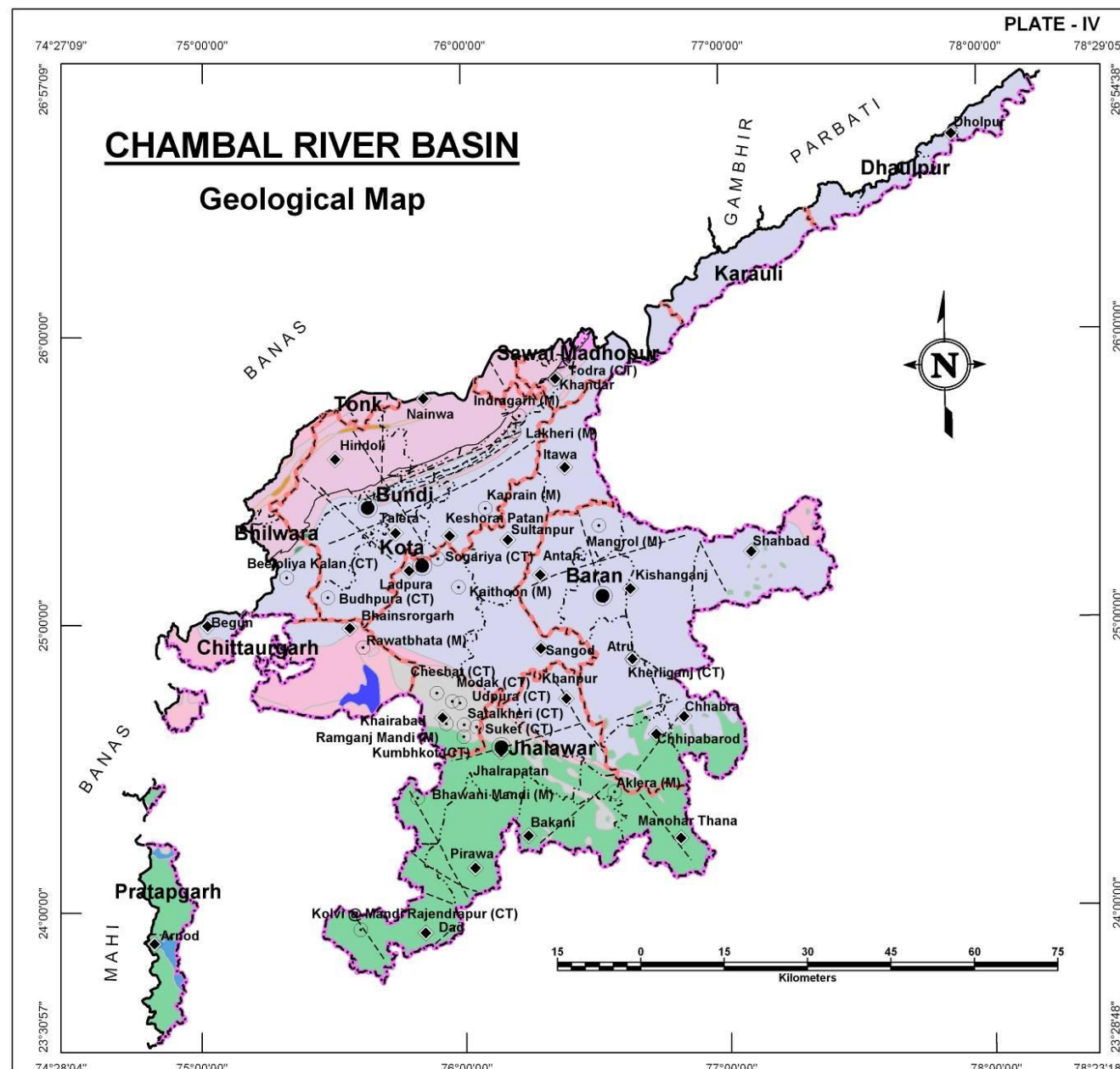
The Chambal river basin is covered predominantly by rocks belonging to the Vindhyan Super-Group and next by Alluvium. Bhilwara (Mangalwar Complex) Super Group rocks and Deccan traps also cover parts of basin.

| Age                                                         | Super Group    | Group- Formation            | Lithology                                                               |
|-------------------------------------------------------------|----------------|-----------------------------|-------------------------------------------------------------------------|
| Sub-recent to Recent                                        | Alluvium       | Alluvium                    | Sand, silt, clay, gravels, Pebbles, kankar                              |
| -----x-----x-----x----- Unconformity-----x-----x-----x----- |                |                             |                                                                         |
| Upper Cretaceous                                            | Deccan Trap    | Malwa                       | Lava flows with Inter-trappean shale, Chert, limestone                  |
| -----x-----x-----x----- Unconformity-----x-----x-----x----- |                |                             |                                                                         |
| Upper Pre-Cambrian to Lower Cambrian                        | Upper Vindhyan | Bhander (Chambal Limestone) | Sandstone, limestone Shale, conglomerate                                |
|                                                             |                | Rewa                        | Sandstone, shale                                                        |
|                                                             |                | Kaimur                      | Sandstone, shale Conglomerates                                          |
| -----x-----x-----x----- Unconformity-----x-----x-----x----- |                |                             |                                                                         |
| Upper Pre- Cambrian to Lower Cambrian                       | Lower Vindhyan | Semri                       | Limestone, shale, Glauconitic beds, Porcellanite, conglomerate          |
| -----x-----x-----x----- Unconformity-----x-----x-----x----- |                |                             |                                                                         |
| Archaean                                                    | Bhilwara       | Mangalwar Complex           | Slate, dolomite, Marble, quartzite, Conglomerate Gneisses and migmatite |

## GEOMORPHOLOGY

| Origin       | Landform Unit          | Description                                                                                                                                                                                                                                 |
|--------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aeolian      | Eolian Plain           | Formed by aeolian activity, with sand dunes of varying height, size, slope. Long stretches of sand sheet. Gently sloping flat to undulating plain, comprised of fine to medium grained sand and silt. Also scattered xerophytic vegetation. |
|              | Sandy Plain            | Formed of aeolian activity, wind-blown sand with gentle sloping to undulating plain, comprising of coarse sand, fine sand, silt and clay.                                                                                                   |
| Denudational | Buried Pediment        | Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.                                                                                                                                             |
|              | Intermontane Valley    | Depression between mountains, generally broad & linear, filled with colluvial deposits.                                                                                                                                                     |
|              | Pediment               | Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.                                                                           |
|              | Dissected Plateau      | Plateau, criss-crossed by fractures forming deep valleys.                                                                                                                                                                                   |
| Fluvial      | Alluvial Fan           | A fan shaped mass of sediment deposit at a point along a Nallah, river where there is a decrease in gradient.                                                                                                                               |
|              | Alluvial Plain         | Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.                                                            |
|              | Alluvial Plain (Sandy) | Flat to gentle undulating plain formed due to fluvial activity, mainly consists of gravels, sand, silt and clay with unconsolidated material of varying lithology, predominantly sand along river.                                          |
|              | Paleochannel           | Mainly buried on abandoned stream/river courses, comprising of coarse textured material of variable sizes.                                                                                                                                  |
|              | Valley Fill            | Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.                                                    |
|              | Ravine                 | Small, narrow, deep, depression, smaller than gorges, larger than gully, usually carved by running water.                                                                                                                                   |
|              | Water logged/ Wetland  | Area submerged in water or area having very shallow water table. So that it submerges in water during rainy season.                                                                                                                         |
| Structural   | Plateau                | Formed over varying lithology with extensive, flat, landscapes, bordered by escarpment on all sides. Essentially formed horizontally layered rocky marked by extensive flat top and steep slopes. It may be criss crossed by lineament.     |
| Hills        |                        | Hills and intervening valleys in hilly areas                                                                                                                                                                                                |







## AQUIFERS

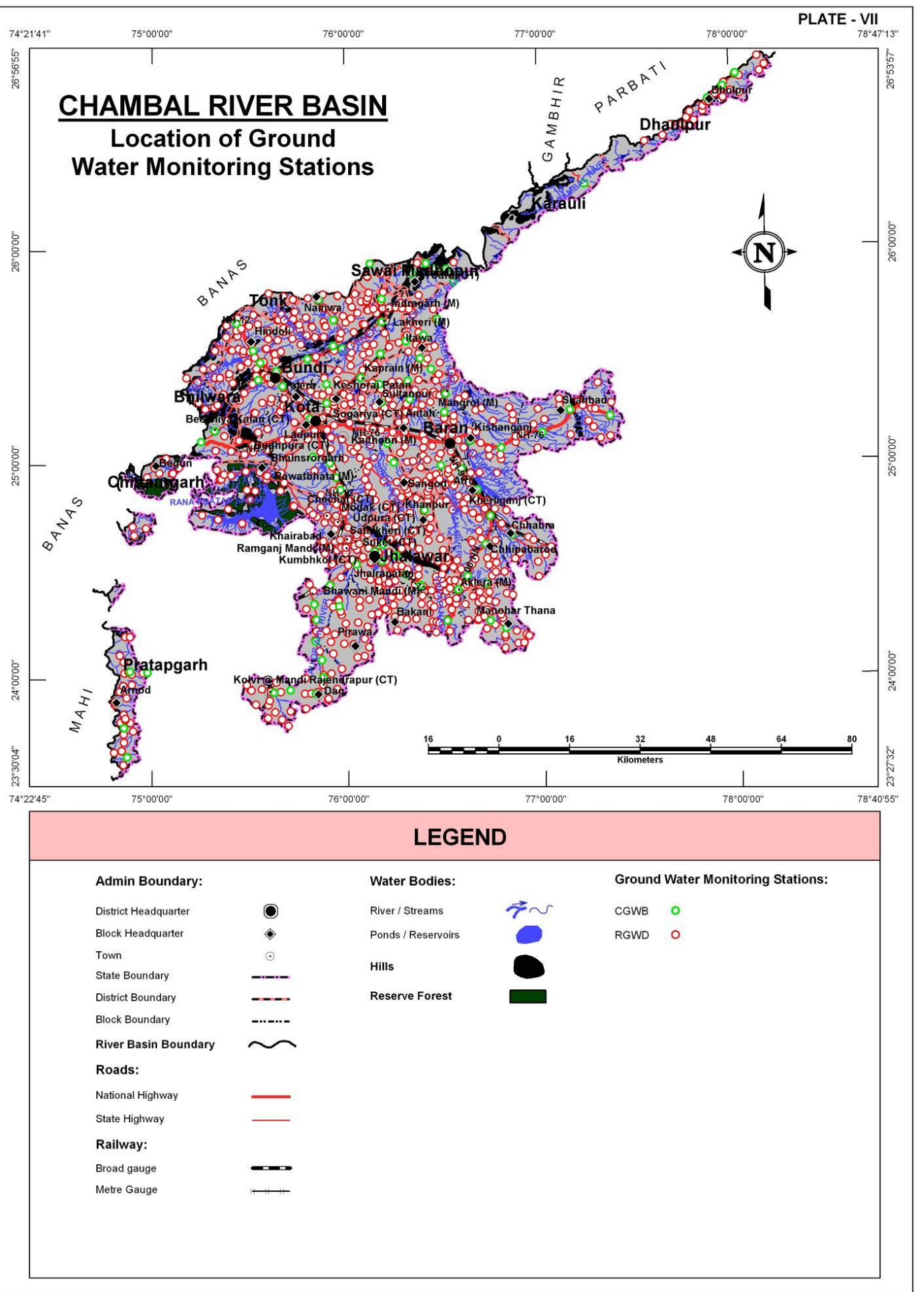
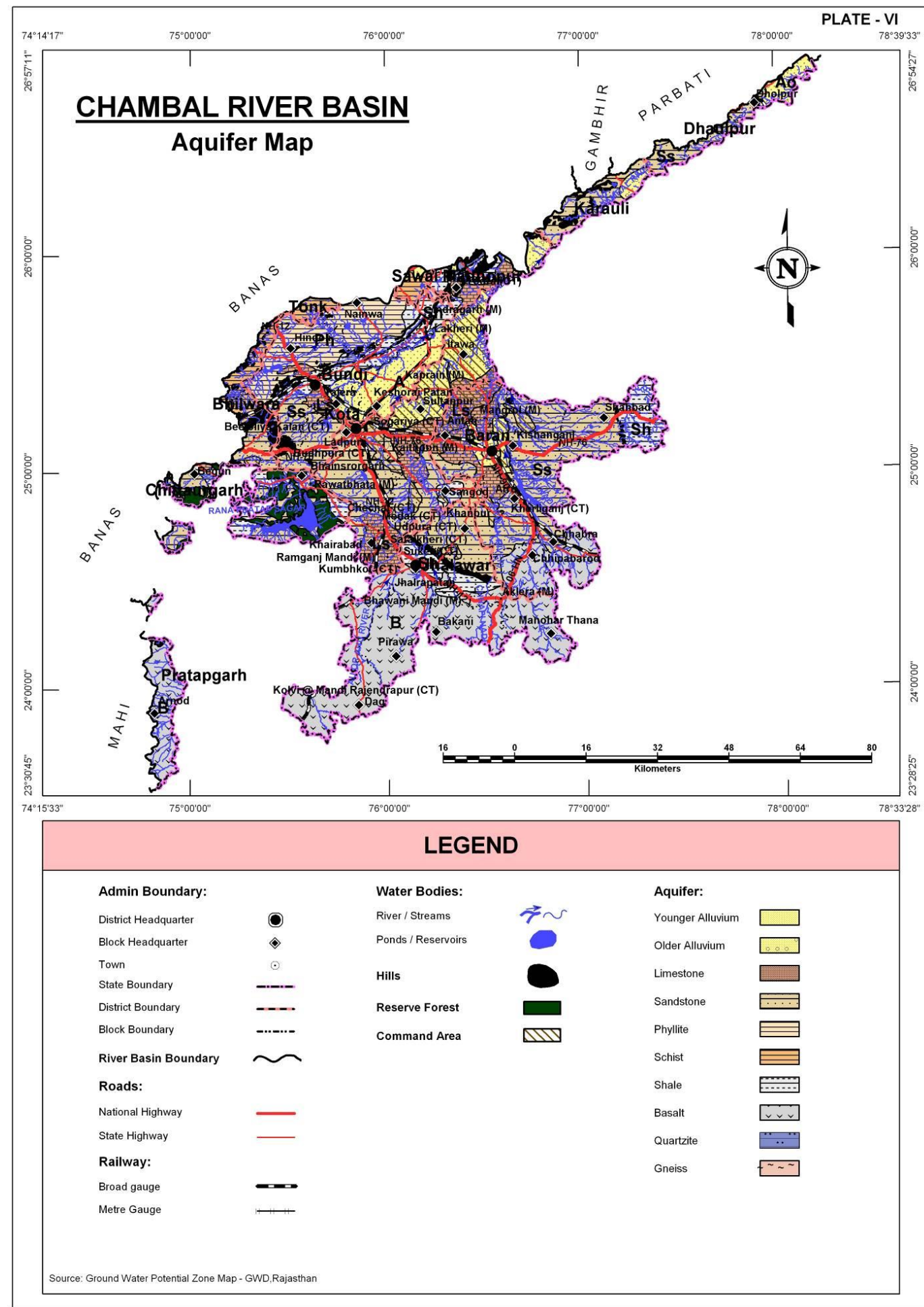
Vindhyan constitute principal aquifers in the basin (sandstones around 35%, followed by shales and limestones around 10% each) mostly in the central parts of the basin. Basaltic aquifers are also significant as they occupy around 22% of the basin area, dominantly present in the southern part of the basin. Alluvial aquifers are present in Kota, Baran and Bundi districts towards northern parts of the basin.

| Aquifer in Potential Zone | Area (sq km)    | % of Basin Area | Description of the unit/Occurrence                                                                            |
|---------------------------|-----------------|-----------------|---------------------------------------------------------------------------------------------------------------|
| Younger Alluvium          | 3,032.2         | 9.7             | It is largely constituted of Aeolian and Fluvial sand, silt, clay, gravel and pebbles in varying proportions. |
| Older Alluvium            | 683.2           | 2.2             | This litho unit comprises of mixture of heterogeneous fine to medium grained sand, silt and kankar.           |
| Limestone                 | 2,892.1         | 9.3             | In general, it is fine to medium grained, grey, red yellowish, pink or buff in colour.                        |
| Vindhyan Sandstone        | 10,140.3        | 32.5            | Fine to medium grained, red colour and compact and at places.                                                 |
| Phyllite                  | 1,963.7         | 6.2             | These include meta sediments and represented by carbonaceous phyllite.                                        |
| Schist                    | 836.4           | 2.7             | Medium to fine grained compact rock. The litho units are soft, friable and have closely spaced cleavage.      |
| Shale                     | 3,097.9         | 9.9             | Grey, light green and purple in colour and mostly splintery in nature.                                        |
| Basalt                    | 6,828.3         | 21.9            | Dark grey, olive green and green colour, compact, vesicular, amygdaloidal and weathered.                      |
| Quartzite                 | 58.6            | 0.2             | Medium to coarse grained and varies from feldspathic grit to sericitic quartzite.                             |
| Gneiss                    | 45.2            | 0.1             | Comprises of porphyritic and non porphyritic gneissic complex.                                                |
| Non Potential Zone        | 1,661.3         | 5.3             | Hills and reserve forests.                                                                                    |
| <b>TOTAL</b>              | <b>31,239.2</b> | <b>100.0</b>    |                                                                                                               |

## LOCATION OF GROUNDWATER MONITORING WELLS

The basin has a well distributed network of large number of groundwater monitoring stations (954) in the basin owned by RGWD (862) and CGWB (92); and 124 wells have been recommended to be added to the existing network to effectively monitor ground water levels and water quality in the basin.

| District Name  | Existing Ground Water Monitoring Stations |            |            | Recommended Additional Ground Water Monitoring Stations |               |
|----------------|-------------------------------------------|------------|------------|---------------------------------------------------------|---------------|
|                | CGWB                                      | RGWD       | Total      | Water Level                                             | Water quality |
| Baran          | 16                                        | 154        | <b>170</b> | -                                                       | 5             |
| Bhilwara       | 2                                         | 17         | <b>19</b>  | -                                                       | -             |
| Bundi          | 18                                        | 206        | <b>224</b> | -                                                       | 3             |
| Chittaurgarh   | -                                         | 33         | <b>33</b>  | -                                                       | 1             |
| Dhaulpur       | 4                                         | 25         | <b>29</b>  | -                                                       | 21            |
| Jhalawar       | 22                                        | 225        | <b>247</b> | -                                                       | 9             |
| Karauli        | 1                                         | -          | <b>1</b>   | 5                                                       | 8             |
| Kota           | 18                                        | 160        | <b>178</b> | 17                                                      | 3             |
| Pratapgarh     | 5                                         | 29         | <b>34</b>  | 37                                                      | 1             |
| Sawai Madhopur | 5                                         | 6          | <b>11</b>  | -                                                       | 14            |
| Tonk           | 1                                         | 7          | <b>8</b>   | -                                                       | -             |
| <b>Total</b>   | <b>92</b>                                 | <b>862</b> | <b>954</b> | <b>59</b>                                               | <b>65</b>     |



## LOCATION OF EXPLORATORY WELLS

## CHAMBAL RIVER BASIN

In all there are 339 exploratory wells present in the basin drilled in the past by RGWD (245) and CGWB (94) that form good basis for delineation of sub-surface aquifer distribution.

| District Name  | Exploratory Wells |            |            |
|----------------|-------------------|------------|------------|
|                | CGWB              | RGWD       | Total      |
| Baran          | 4                 | 43         | 47         |
| Bhilwara       | -                 | 3          | 3          |
| Bundi          | 26                | 57         | 83         |
| Chittaurgarh   | 2                 | 4          | 6          |
| Dhaulpur       | 1                 | 8          | 9          |
| Jhalawar       | 17                | 57         | 74         |
| Karauli        | 2                 | -          | 2          |
| Kota           | 34                | 47         | 81         |
| Pratapgarh     | -                 | 3          | 3          |
| Sawai Madhopur | 8                 | 18         | 26         |
| Tonk           | -                 | 5          | 5          |
| <b>Total</b>   | <b>94</b>         | <b>245</b> | <b>339</b> |

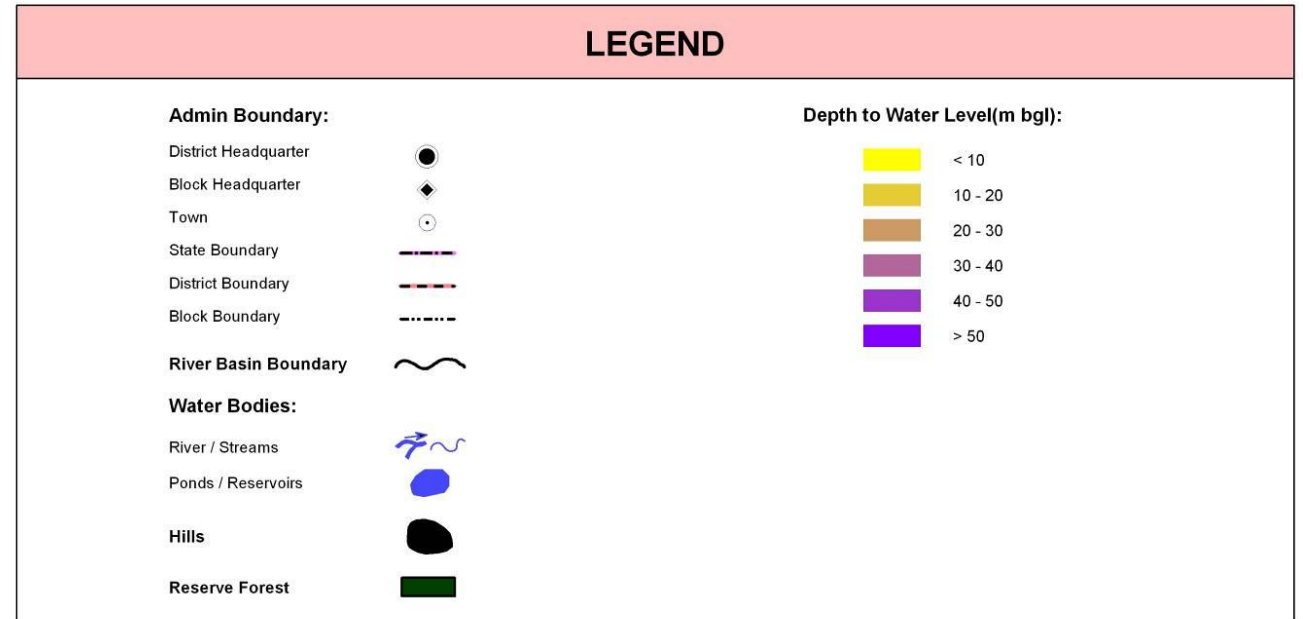
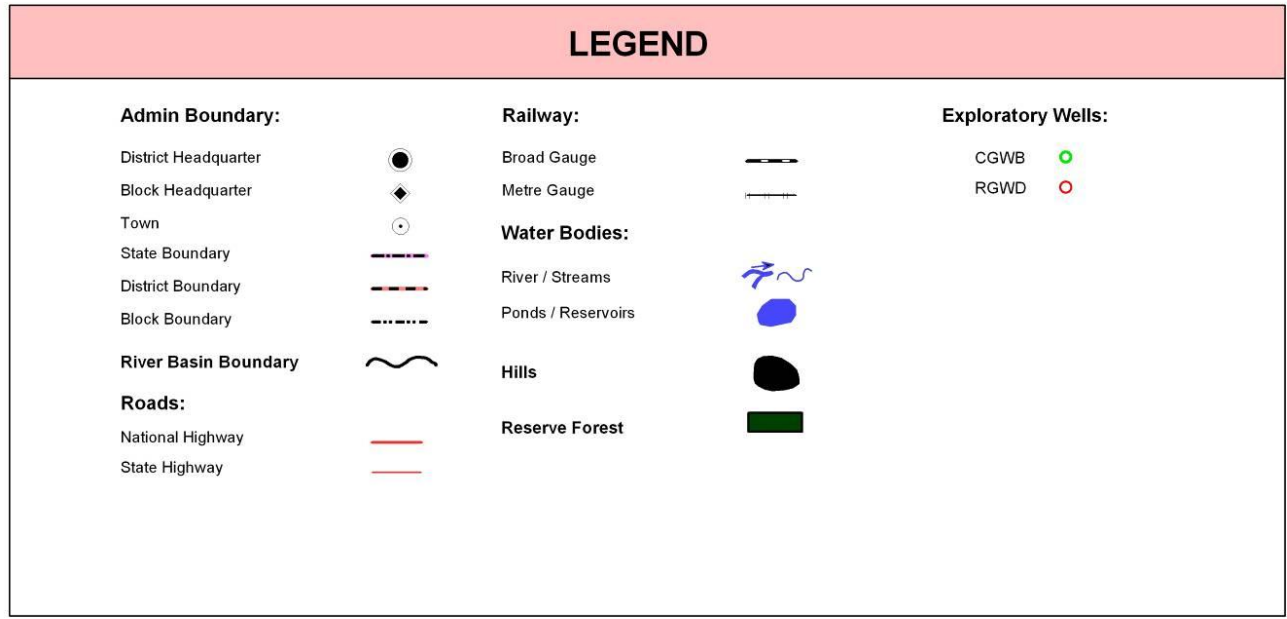
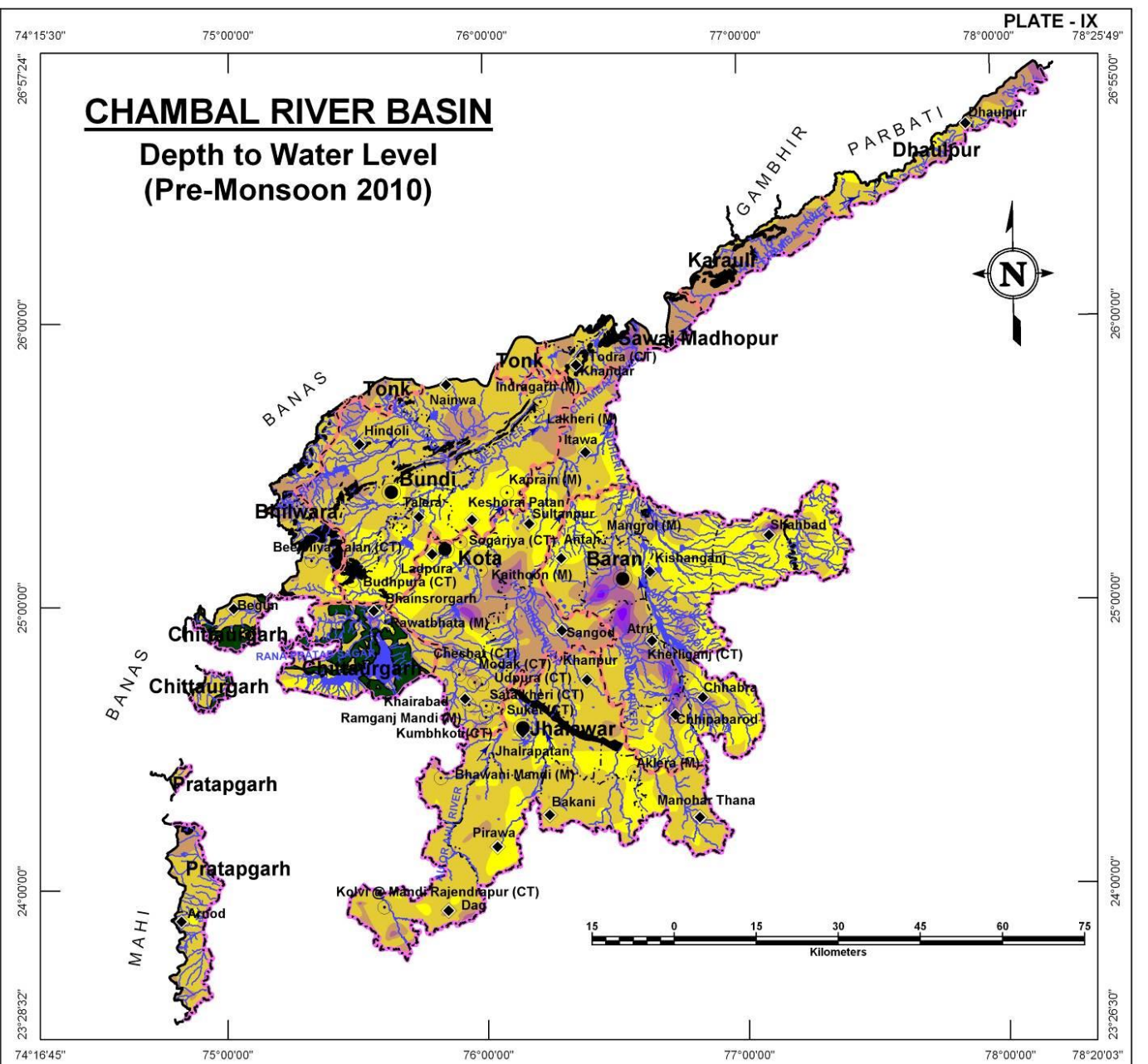
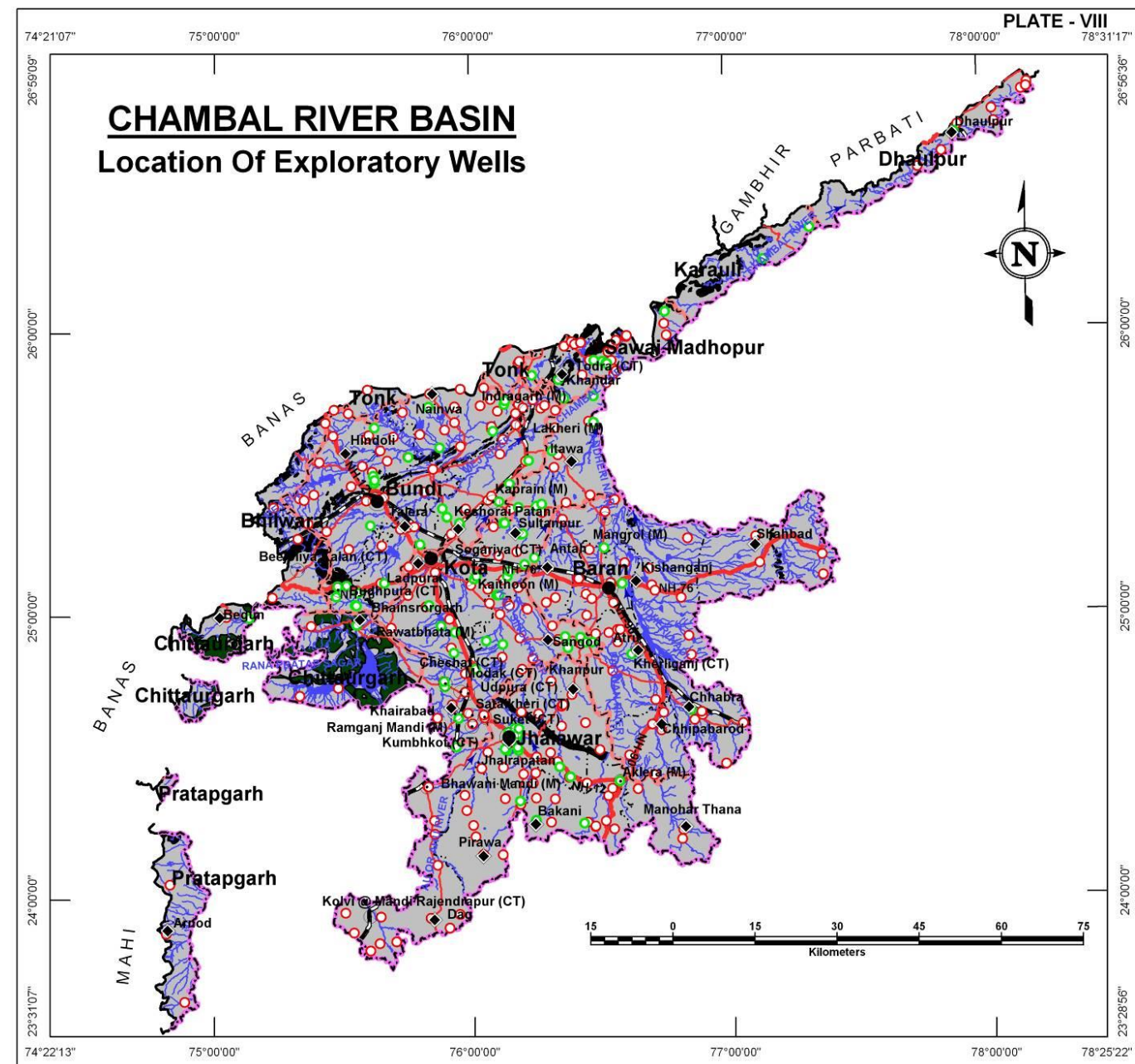
## DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

The general depth to water level in the basin ranges from 10 to 30 meters below ground level, as seen all over the basin except the central part in Kota-Baran-Chhabra region. The deeper water levels in the region mentioned above reaches 50m bgl in some areas.

| Depth to water level<br>(m bgl)<br>Pre Monsoon - 2010 | District wise area coverage (sq km)* |              |                |                |              |                |              |                |              |                   |              | Total Area<br>(sq km) |
|-------------------------------------------------------|--------------------------------------|--------------|----------------|----------------|--------------|----------------|--------------|----------------|--------------|-------------------|--------------|-----------------------|
|                                                       | Baran                                | Bhilwara     | Bundi          | Chittaurgarh   | Dhaulpur     | Jhalawar       | Karauli      | Kota           | Pratapgarh   | Sawai<br>Madhopur | Tonk         |                       |
| < 10                                                  | 3,111.0                              | 45.6         | 1,457.9        | 241.2          | 77.7         | 1,058.4        | -            | 1,203.1        | 22.4         | -                 | -            | 7,217.3               |
| 10 - 20                                               | 2,672.3                              | 583.5        | 3,217.6        | 983.0          | 482.1        | 4,860.6        | 63.5         | 2,492.3        | 709.3        | 383.2             | 379.8        | 16,827.2              |
| 20 - 30                                               | 707.3                                | 191.5        | 747.9          | 92.5           | 229.4        | 261.6          | 638.4        | 1,258.1        | 142.9        | 384.3             | 66.2         | 4,720.1               |
| 30 - 40                                               | 334.0                                | -            | 2.0            | 9.0            | 55.7         | 24.5           | 8.4          | 146.7          | -            | 63.5              | -            | 643.8                 |
| 40 - 50                                               | 116.0                                | -            | -              | 2.0            | -            | 0.2            | -            | 22.1           | -            | 3.6               | -            | 143.9                 |
| > 50                                                  | 25.5                                 | -            | -              | -              | -            | -              | -            | -              | -            | -                 | -            | 25.5                  |
| <b>Total</b>                                          | <b>6,966.1</b>                       | <b>820.6</b> | <b>5,425.4</b> | <b>1,327.7</b> | <b>844.9</b> | <b>6,205.3</b> | <b>710.3</b> | <b>5,122.3</b> | <b>874.6</b> | <b>834.6</b>      | <b>446.0</b> | <b>29,577.8</b>       |

\* The area covered in the derived maps is less than the total basin area since the hills have been excluded from interpolation/contouring.





## WATER TABLE ELEVATION (PRE MONSOON 2010)

## CHAMBAL RIVER BASIN

Water table elevation map Plate –X, shows large variation following the trends in topography which is marked by hills and low plains. It varies from less than 140m amsl (in Dhaulpur district) to about 520m amsl (in Pratapgarh district). Predominant elevation range is between 220 to 260m amsl range wherein, 27% of the area falls under this region. Most of the remaining of the area has water table between 260 to 440m amsl, spread over the Baran, Bhilwara, Bundi, Chittaurgarh, Jhalawar, Kota districts. Dhaulpur district has generally lowest water table elevation values in the basin.

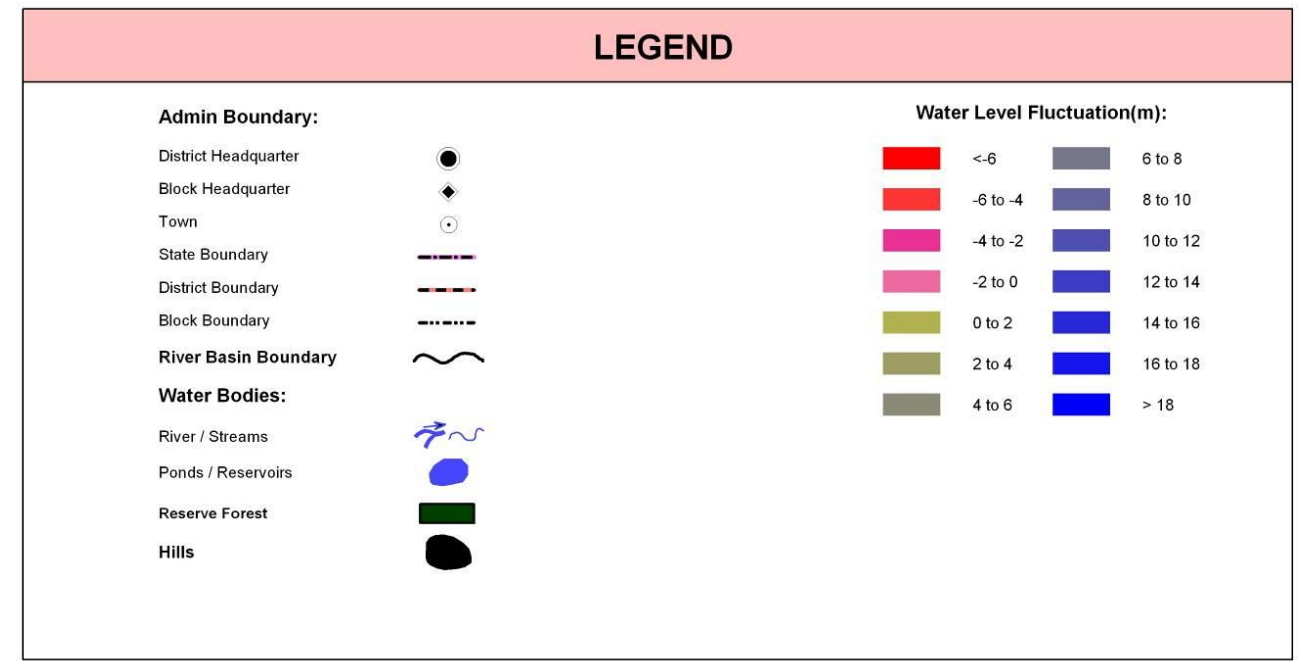
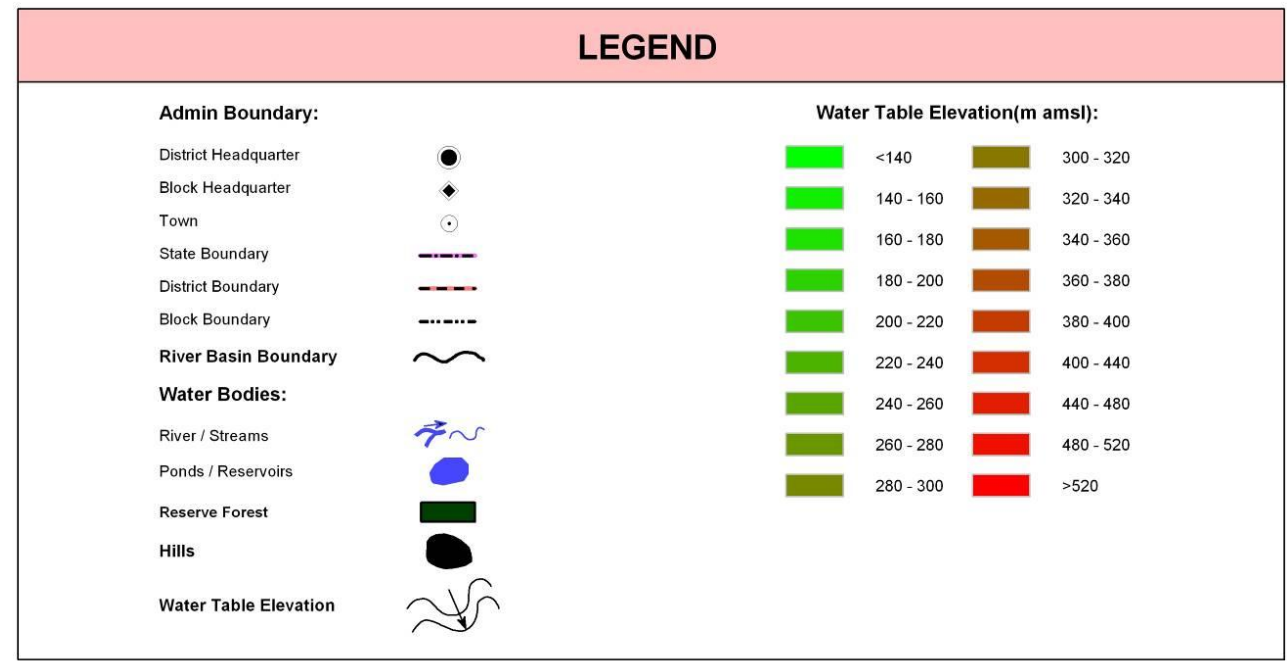
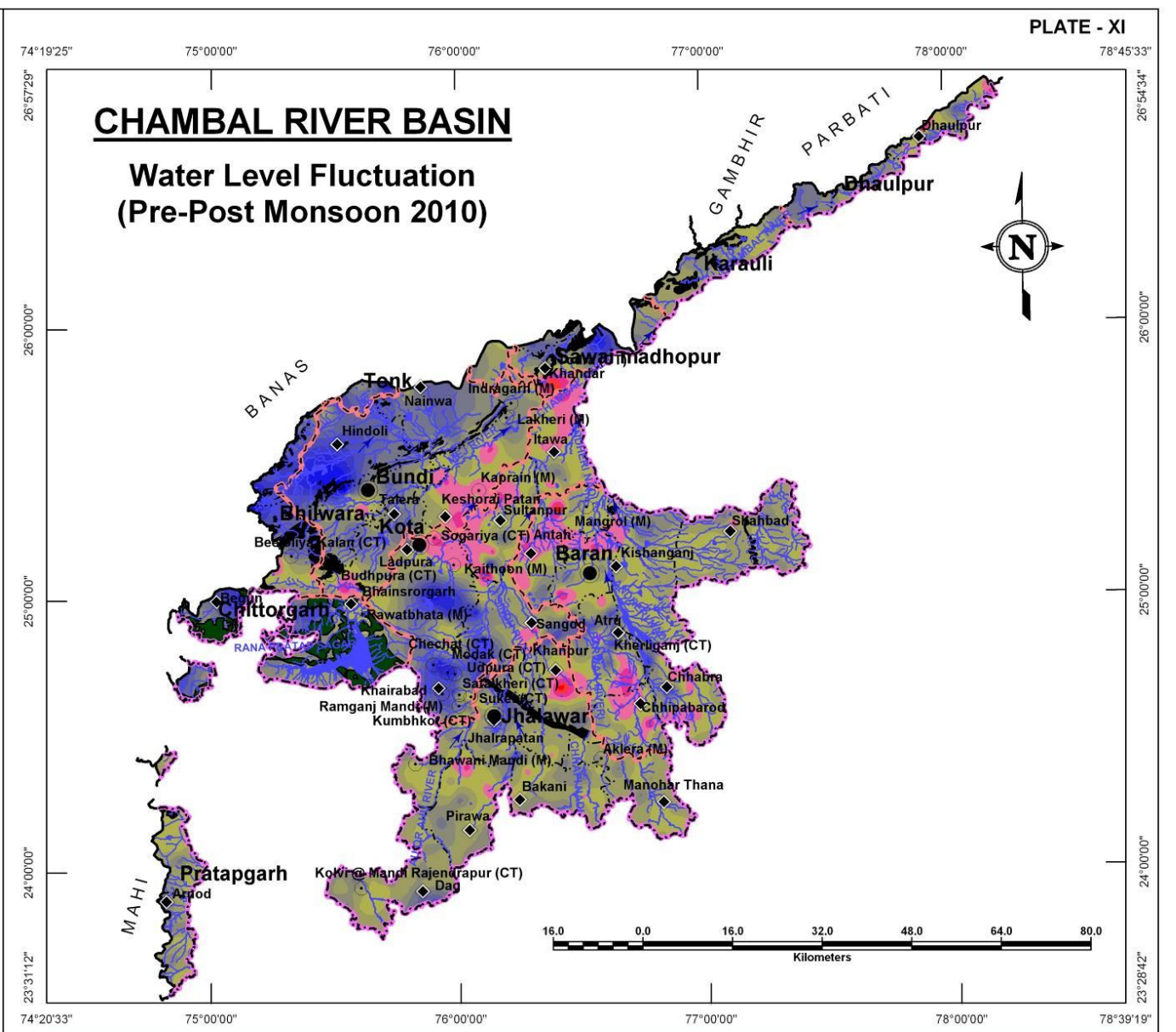
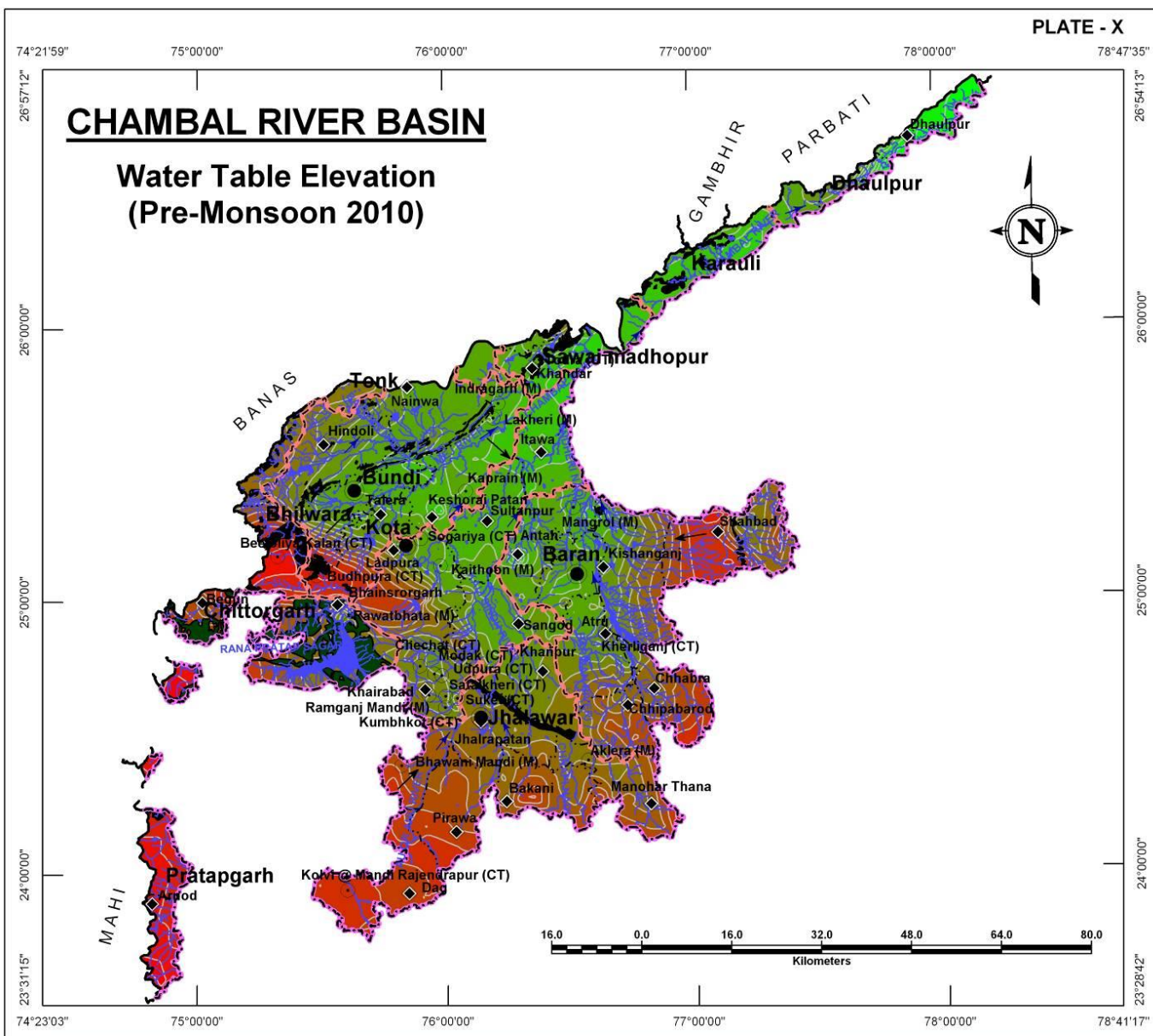
| Depth to water table (m bgl)<br>Pre Monsoon - 2010 | District wise area coverage (sq km) |              |                |                |              |                |              |                |              |                |              | Total Area<br>(sq km) |
|----------------------------------------------------|-------------------------------------|--------------|----------------|----------------|--------------|----------------|--------------|----------------|--------------|----------------|--------------|-----------------------|
|                                                    | Baran                               | Bhilwara     | Bundi          | Chittaurgarh   | Dhaulpur     | Jhalawar       | Karauli      | Kota           | Pratapgarh   | Sawai Madhopur | Tonk         |                       |
| < 140                                              | -                                   | -            | -              | -              | 246.3        | -              | -            | -              | -            | -              | -            | 246.3                 |
| 140 - 160                                          | -                                   | -            | -              | -              | 161.9        | -              | -            | -              | -            | -              | -            | 161.9                 |
| 160 - 180                                          | 3.2                                 | -            | -              | -              | 33.6         | -              | 4.3          | -              | -            | -              | -            | 41.1                  |
| 180 - 200                                          | 11.9                                | -            | 11.3           | -              | 33.6         | -              | 149.8        | 99.2           | -            | 178.7          | -            | 484.5                 |
| 200 - 220                                          | 57.4                                | -            | 293.2          | -              | 26.9         | -              | 478.1        | 608.4          | -            | 352.0          | -            | 1,816.0               |
| 220 - 240                                          | 659.9                               | -            | 1,277.5        | -              | 94.6         | -              | 71.9         | 1,125.9        | -            | 169.2          | 34.5         | 3,433.5               |
| 240 - 260                                          | 1,308.6                             | -            | 1,730.7        | -              | 166.0        | 8.6            | 6.2          | 935.3          | -            | 103.4          | 223.2        | 4,482.0               |
| 260 - 280                                          | 604.5                               | -            | 627.2          | 10.0           | 69.3         | 256.1          | -            | 518.1          | -            | 12.0           | -            | 2,097.2               |
| 280 - 300                                          | 508.4                               | -            | 476.9          | 50.5           | 12.7         | 450.8          | -            | 705.8          | -            | 11.2           | 73.7         | 2,290.0               |
| 300 - 320                                          | 682.5                               | 98.7         | 375.9          | 374.1          | -            | 823.9          | -            | 468.3          | -            | 7.5            | 107.0        | 2,937.9               |
| 320 - 340                                          | 975.6                               | 203.4        | 108.6          | 159.2          | -            | 1,044.0        | -            | 192.6          | -            | 0.6            | 7.6          | 2,691.6               |
| 340 - 360                                          | 730.4                               | 62.8         | 71.8           | 204.9          | -            | 1,101.8        | -            | 147.6          | -            | -              | -            | 2,319.3               |
| 360 - 380                                          | 593.1                               | 52.4         | 67.7           | 208.0          | -            | 837.7          | -            | 153.6          | -            | -              | -            | 1,912.5               |
| 380 - 400                                          | 319.8                               | 32.1         | 81.2           | 137.6          | -            | 849.5          | -            | 159.1          | -            | -              | -            | 1,579.3               |
| 400 - 440                                          | 479.0                               | 63.8         | 168.5          | 55.2           | -            | 832.9          | -            | 8.4            | 6.8          | -              | -            | 1,614.6               |
| 440 - 480                                          | 31.8                                | 166.4        | 122.8          | 10.2           | -            | -              | -            | -              | 343.4        | -              | -            | 674.6                 |
| 480 - 520                                          | -                                   | 141.0        | 12.1           | 56.6           | -            | -              | -            | -              | 472.7        | -              | -            | 682.4                 |
| > 520                                              | -                                   | -            | -              | 61.4           | -            | -              | -            | -              | 51.7         | -              | -            | 113.1                 |
| <b>Total</b>                                       | <b>6,966.1</b>                      | <b>820.6</b> | <b>5,425.4</b> | <b>1,327.7</b> | <b>844.9</b> | <b>6,205.3</b> | <b>710.3</b> | <b>5,122.3</b> | <b>874.6</b> | <b>834.6</b>   | <b>446.0</b> | <b>29,577.8</b>       |

## WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

The Basin area is a mixture of hardrock aquifers and those formed within alluvium aquifers. Perusal of Plate XI reveals interesting correlation wherein the hardrock areas have shown positive fluctuation (i.e., rise in ground water levels in post monsoon period) to the tune of 2 to even 18m (Bundi, Kota, Chittaurgarh, and Sawai Madhopur being prominent), the alluvial areas of Bundi, Kota and Baran have shown significant areas having negative fluctuation (fall in water level in post monsoon period) between 0 to 4 m reaching even upto 6m in some areas.

| District Name  | District wise area coverage (sq km) within fluctuation range (m) |             |              |                |                |                |                |                |                |                |              |              |              |             | Total Area<br>(sq km) |
|----------------|------------------------------------------------------------------|-------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|--------------|--------------|--------------|-------------|-----------------------|
|                | < -6                                                             | -6 to -4    | -4 to -2     | -2 to 0        | 0 to 2         | 2 to 4         | 4 to 6         | 6 to 8         | 8 to 10        | 10 to 12       | 12 to 14     | 14 to 16     | 16 to 18     | > 18        |                       |
| Baran          | 3.5                                                              | 9.4         | 114.1        | 1,043.6        | 2,631.7        | 2,122.5        | 727.2          | 214.0          | 63.2           | 30.5           | 5.9          | 0.5          | -            | -           | 6,966.1               |
| Bhilwara       | -                                                                | -           | 1.3          | 16.8           | 109.6          | 97.1           | 49.2           | 79.8           | 168.6          | 135.3          | 112.5        | 35.1         | 13.7         | 1.6         | 820.6                 |
| Bundi          | -                                                                | -           | 57.9         | 448.1          | 564.9          | 793.1          | 1,185.3        | 996.5          | 677.2          | 246.2          | 181.4        | 144.0        | 116.5        | 14.3        | 5,425.4               |
| Chittaurgarh   | -                                                                | -           | -            | -              | 84.8           | 298.2          | 417.1          | 185.6          | 167.8          | 109.5          | 48.5         | 16.2         | -            | -           | 1,327.7               |
| Dhaulpur       | 0.9                                                              | 2.1         | 8.7          | 34.5           | 101.3          | 245.7          | 233.8          | 189.6          | 27.8           | 0.5            | -            | -            | -            | -           | 844.9                 |
| Jhalawar       | 4.1                                                              | 14.0        | 45.2         | 164.8          | 1,534.7        | 2,746.3        | 1,257.2        | 307.7          | 100.9          | 24.6           | 4.3          | 1.3          | 0.2          | -           | 6,205.3               |
| Karauli        | -                                                                | -           | -            | -              | 112.5          | 407.4          | 178.7          | 11.7           | -              | -              | -            | -            | -            | -           | 710.3                 |
| Kota           | 2.3                                                              | 6.9         | 88.1         | 782.1          | 1,261.1        | 900.6          | 607.8          | 470.7          | 343.8          | 356.0          | 185.0        | 103.3        | 14.6         | -           | 5,122.3               |
| Pratapgarh     | -                                                                | -           | -            | -              | 374.0          | 326.3          | 167.1          | 7.2            | -              | -              | -            | -            | -            | -           | 874.6                 |
| Sawai Madhopur | 2.0                                                              | 24.3        | 40.7         | 41.6           | 47.2           | 175.5          | 175.0          | 123.1          | 75.7           | 63.9           | 39.6         | 22.2         | 3.8          | -           | 834.6                 |
| Tonk           | -                                                                | -           | -            | -              | 16.4           | 170.5          | 71.5           | 57.6           | 71.7           | 37.8           | 10.6         | 6.3          | 3.6          | -           | 446.0                 |
| <b>Total</b>   | <b>12.8</b>                                                      | <b>56.7</b> | <b>356.0</b> | <b>2,531.5</b> | <b>6,838.2</b> | <b>8,283.2</b> | <b>5,069.9</b> | <b>2,643.5</b> | <b>1,696.7</b> | <b>1,004.3</b> | <b>587.8</b> | <b>328.9</b> | <b>152.4</b> | <b>15.9</b> | <b>29,577.8</b>       |







## ELECTRICAL CONDUCTIVITY DISTRIBUTION

## CHAMBAL RIVER BASIN

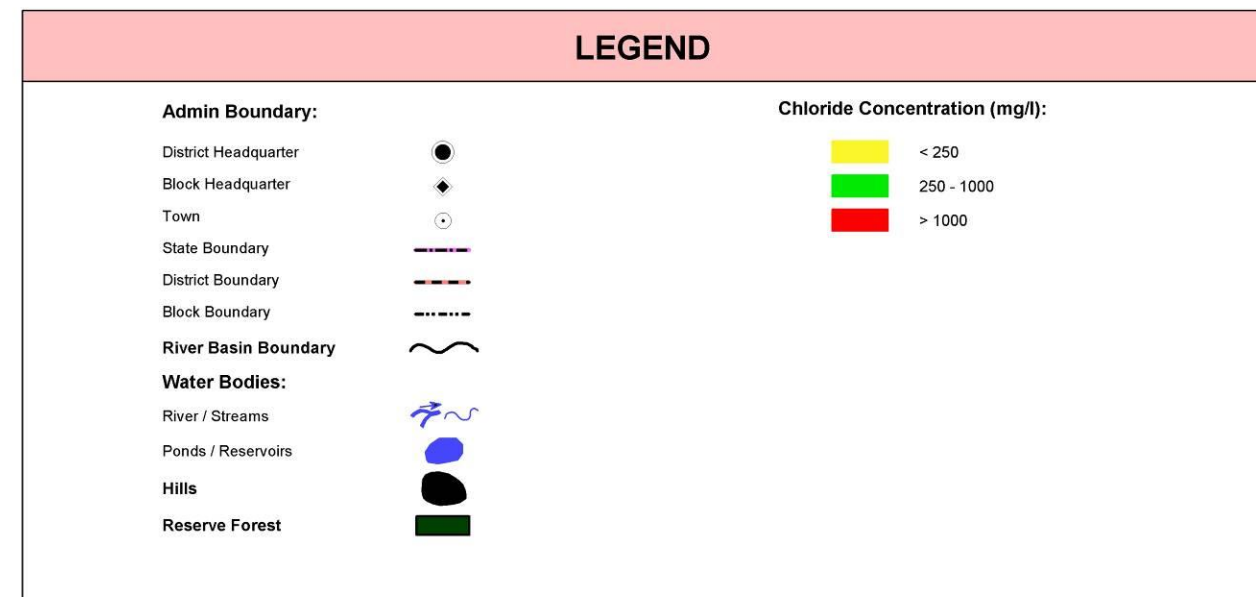
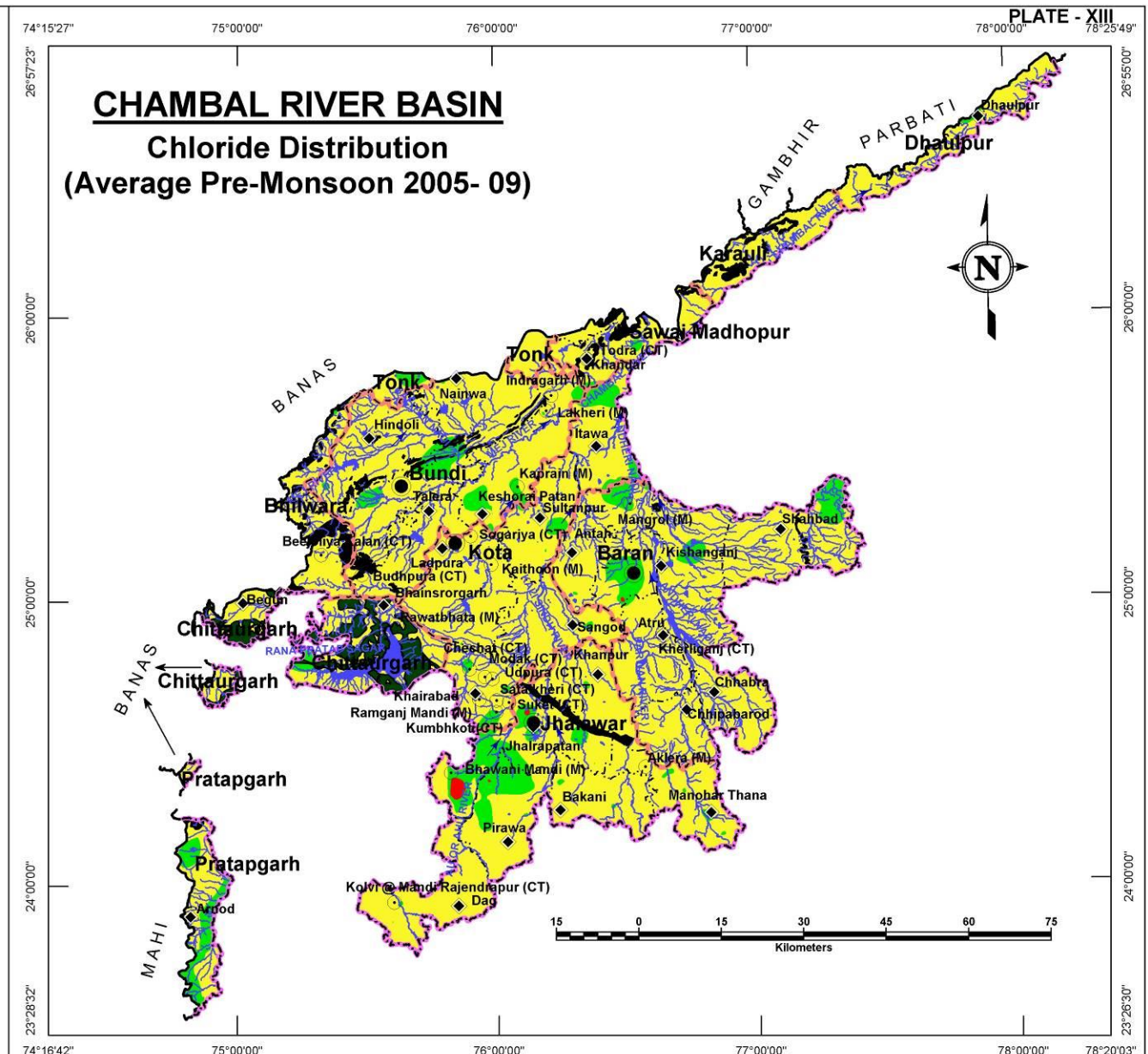
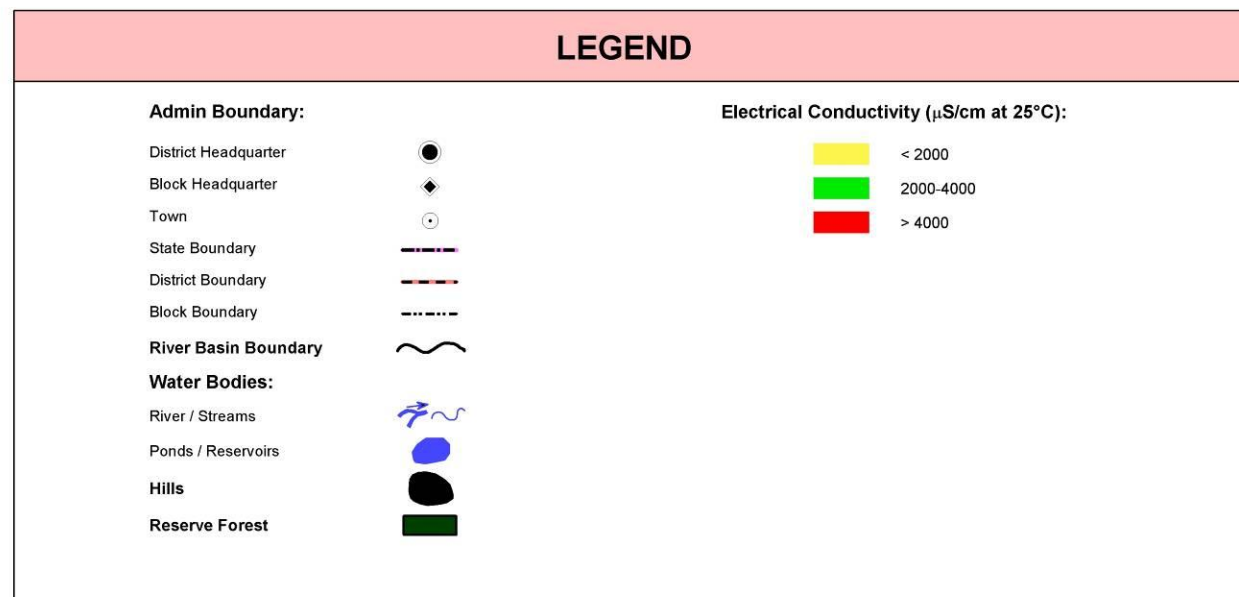
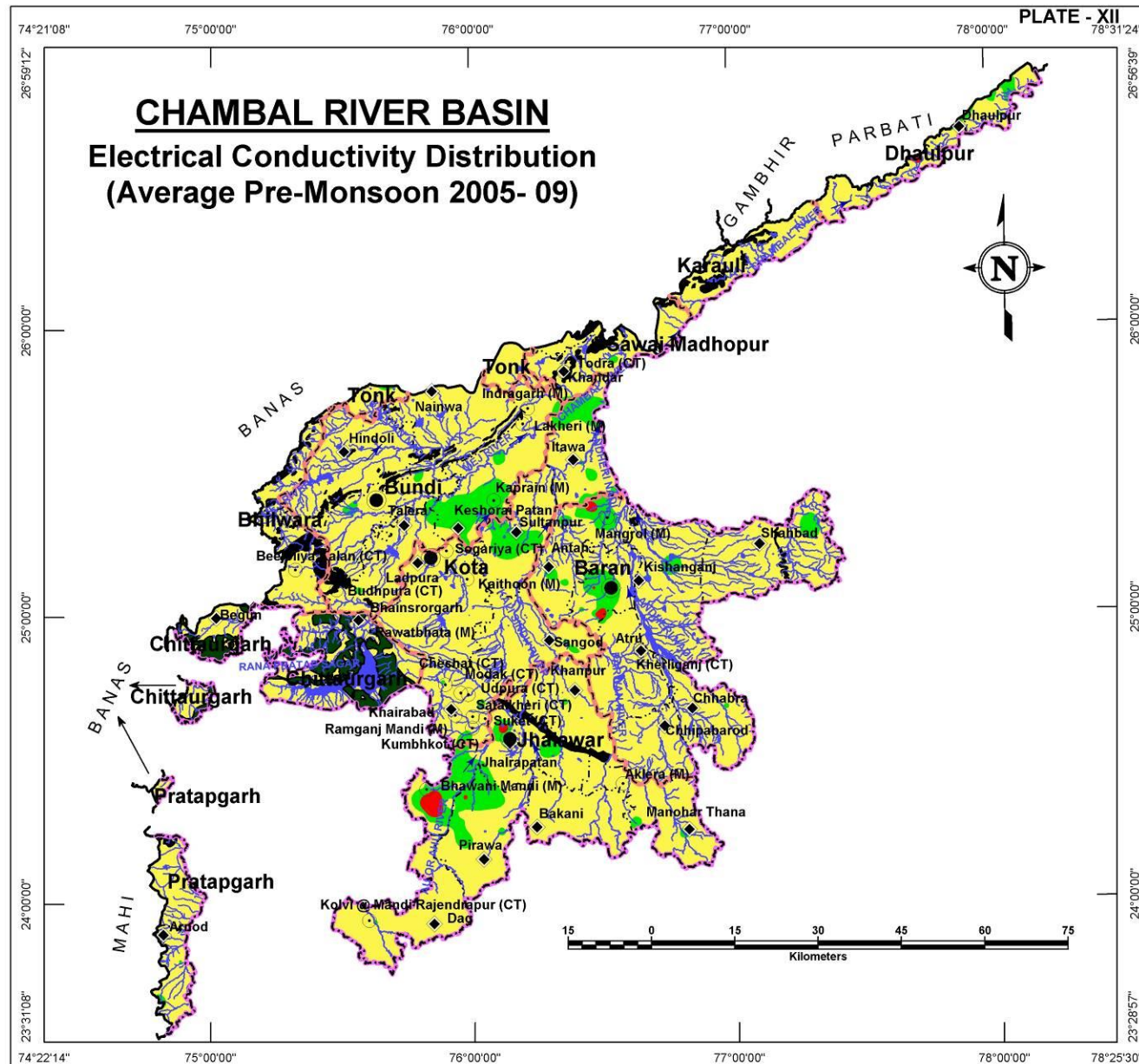
The Electrical Conductivity (at 25°C) distribution map is presented in Plate – XII. Interestingly, 92% of the basin area falls under the < 2000 mg/l category which is suitable for domestic and other purposes from EC point of view. Nearly 8% of the area falls in the 2000 to 4000 mg/l range and negligible area is seen to have over > 4000 mg/l of EC in ground water. Most of the low EC areas correspond to hard rock aquifers whereas the moderately high EC regions correspond to alluvial aquifers in plain areas. The small patch of high EC southwest of Jhalwar is however formed in Basaltic aquifer which may have local reasons for this.

| Electrical Conductivity Ranges<br>(μS/cm at 25°C)<br>(Ave. of years 2005-09) | District wise area coverage (sq km) |       |          |       |         |       |              |       |          |       |          |       |         |       |         |       |            |       |                |       |       |       | Total Area<br>(sq km) |
|------------------------------------------------------------------------------|-------------------------------------|-------|----------|-------|---------|-------|--------------|-------|----------|-------|----------|-------|---------|-------|---------|-------|------------|-------|----------------|-------|-------|-------|-----------------------|
|                                                                              | Baran                               |       | Bhilwara |       | Bundi   |       | Chittaurgarh |       | Dhaulpur |       | Jhalawar |       | Karauli |       | Kota    |       | Pratapgarh |       | Sawai Madhopur |       | Tonk  |       |                       |
|                                                                              | Area                                | % age | Area     | % age | Area    | % age | Area         | % age | Area     | % age | Area     | % age | Area    | % age | Area    | % age | Area       | % age | Area           | % age | Area  | % age |                       |
| < 2000                                                                       | 6,458.3                             | 92.7  | 820.6    | 100.0 | 4,915.3 | 90.6  | 1,327.7      | 100.0 | 765.7    | 90.6  | 5,411.2  | 87.2  | 710.3   | 100.0 | 4,622.1 | 90.2  | 862.8      | 98.6  | 816.3          | 97.8  | 437.7 | 98.1  | 27,148.0              |
| 2000-4000                                                                    | 478.6                               | 6.9   | -        | -     | 510.1   | 9.4   | -            | -     | 65.8     | 7.8   | 712.2    | 11.5  | -       | -     | 500.1   | 9.8   | 11.8       | 1.4   | 18.3           | 2.2   | 8.3   | 1.9   | 2,305.2               |
| > 4000                                                                       | 29.2                                | 0.4   | -        | -     | -       | -     | -            | -     | 13.4     | 1.6   | 81.9     | 1.3   | -       | -     | 0.1     | -     | -          | -     | -              | -     | -     | -     | 124.6                 |
| Total                                                                        | 6,966.1                             | 100.0 | 820.6    | 100.0 | 5,425.4 | 100.0 | 1,327.7      | 100.0 | 844.9    | 100.0 | 6,205.3  | 100.0 | 710.3   | 100.0 | 5,122.3 | 100.0 | 874.6      | 100.0 | 834.6          | 100.0 | 446.0 | 100.0 | 29,577.8              |

## CHLORIDE DISTRIBUTION

Yellow coloured area in Plate – XIII represents the areas of low (< 250 mg/l) concentration of Chloride in ground water. This range is most widespread and almost entire basin shows low Chloride concentrations making the ground water suitable for all purposes. There is a small patch in the southwest of Jhalwar where apparently a single well had shown high concentration of Chloride (>1000 mg/l). Majority of patches of moderately high Chloride concentration are found in alluvial aquifers and the hardrock aquifers seem to contain good quality of ground water.

| Chloride Ranges<br>(mg/l)<br>(Ave. of years 2005-09) | District wise area coverage (sq km) |       |          |       |         |       |              |       |          |       |          |       |         |       |         |       |            |       |                |       |       |       | Total Area<br>(sq km) |
|------------------------------------------------------|-------------------------------------|-------|----------|-------|---------|-------|--------------|-------|----------|-------|----------|-------|---------|-------|---------|-------|------------|-------|----------------|-------|-------|-------|-----------------------|
|                                                      | Baran                               |       | Bhilwara |       | Bundi   |       | Chittaurgarh |       | Dhaulpur |       | Jhalawar |       | Karauli |       | Kota    |       | Pratapgarh |       | Sawai Madhopur |       | Tonk  |       |                       |
|                                                      | Area                                | % age | Area     | % age | Area    | % age | Area         | % age | Area     | % age | Area     | % age | Area    | % age | Area    | % age | Area       | % age | Area           | % age | Area  | % age |                       |
| < 250                                                | 6,347.4                             | 91.1  | 811.4    | 98.9  | 4,973.4 | 91.7  | 1,267.2      | 95.4  | 820.5    | 97.1  | 5,294.6  | 85.3  | 710.3   | 100.0 | 4,761.2 | 92.9  | 537.0      | 61.4  | 797.7          | 95.6  | 390.9 | 87.7  | 26,711.6              |
| 250 - 1000                                           | 614.6                               | 8.8   | 9.2      | 1.1   | 452.0   | 8.3   | 60.5         | 4.6   | 24.4     | 2.9   | 856.1    | 13.8  | -       | -     | 361.1   | 7.1   | 337.6      | 38.6  | 36.9           | 4.4   | 55.1  | 12.3  | 2,807.5               |
| > 1000                                               | 4.1                                 | 0.1   | -        | -     | -       | -     | -            | -     | -        | -     | 54.6     | 0.9   | -       | -     | -       | -     | -          | -     | -              | -     | -     | -     | 58.7                  |
| Total                                                | 6,966.1                             | 100.0 | 820.6    | 100.0 | 5,425.4 | 100.0 | 1,327.7      | 100.0 | 844.9    | 100.0 | 6,205.3  | 100.0 | 710.3   | 100.0 | 5,122.3 | 100.0 | 874.6      | 100.0 | 834.6          | 100.0 | 446.0 | 100.0 | 29,577.8              |





## FLUORIDE DISTRIBUTION

## CHAMBAL RIVER BASIN

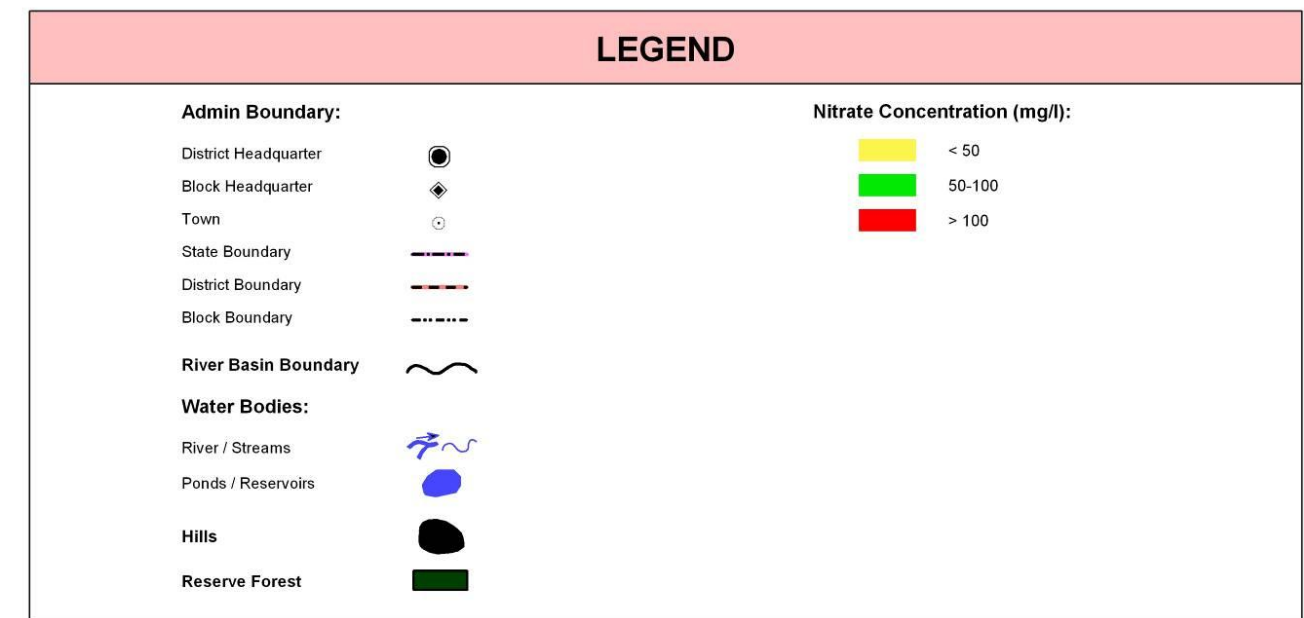
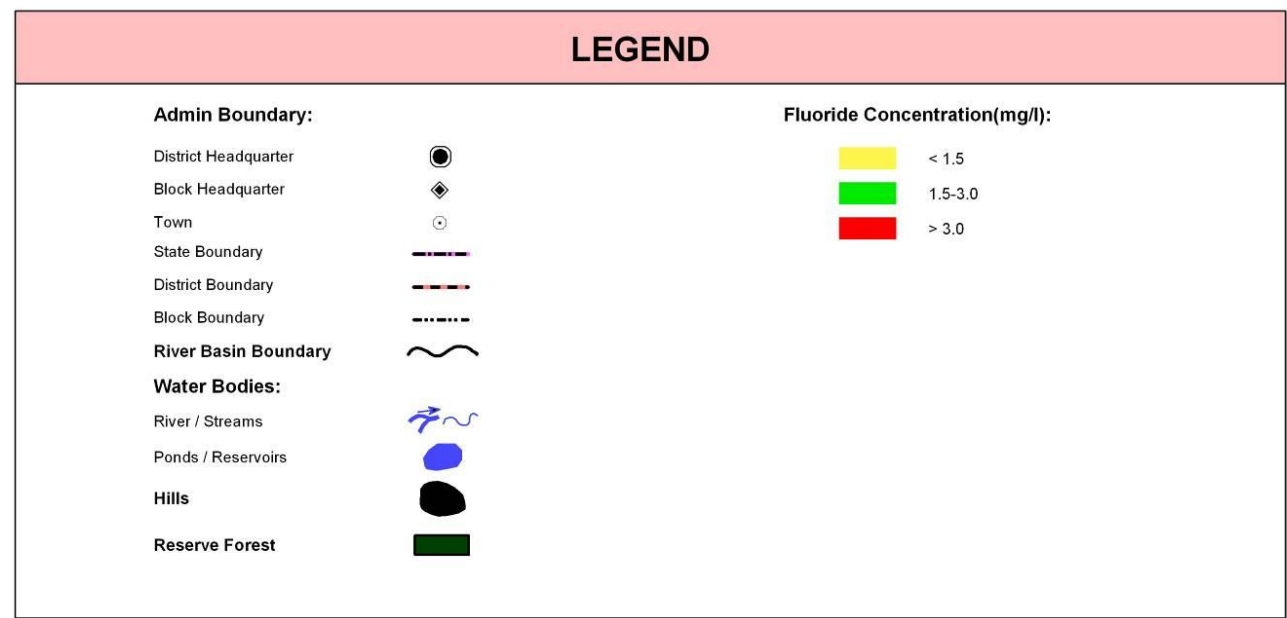
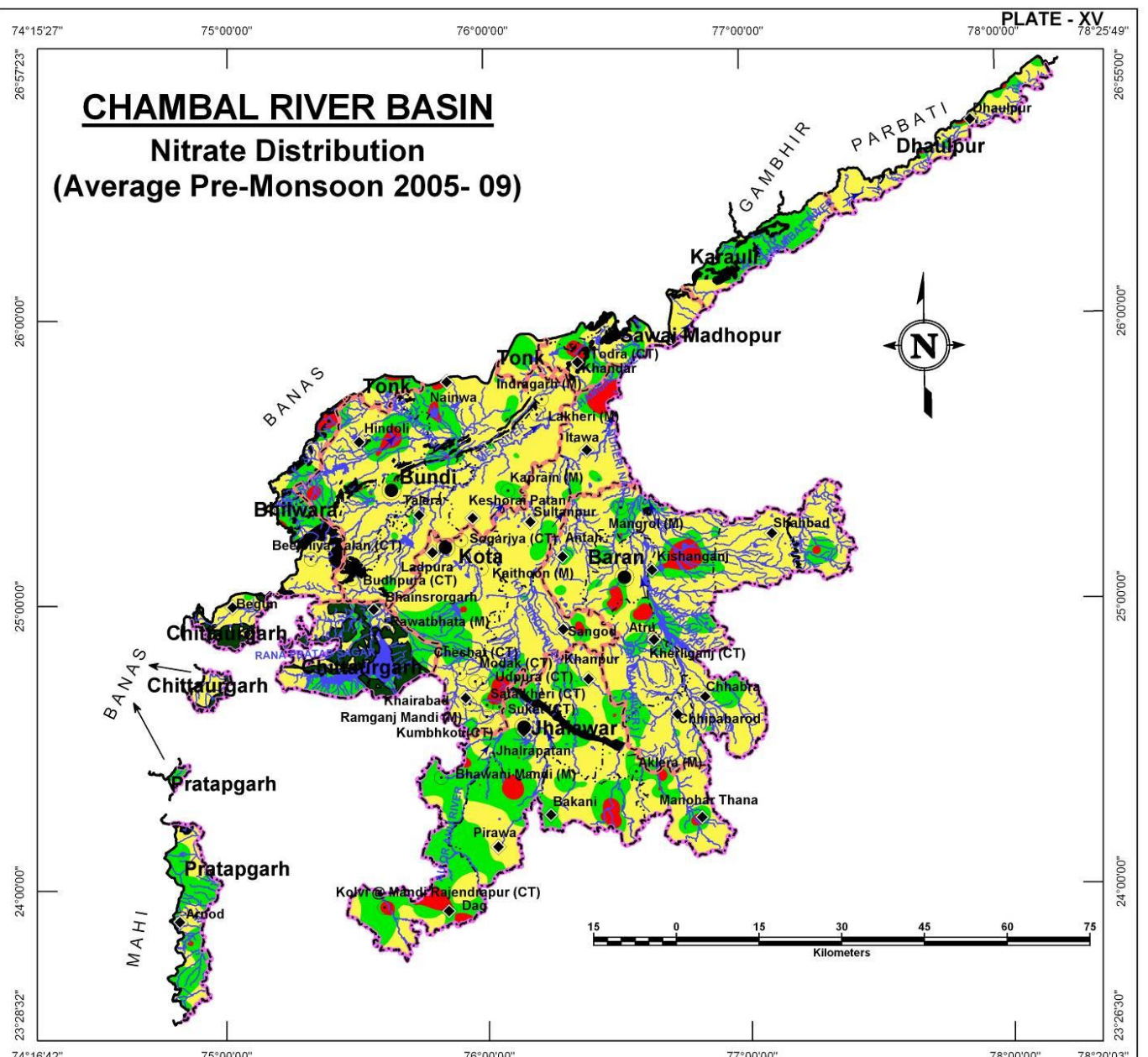
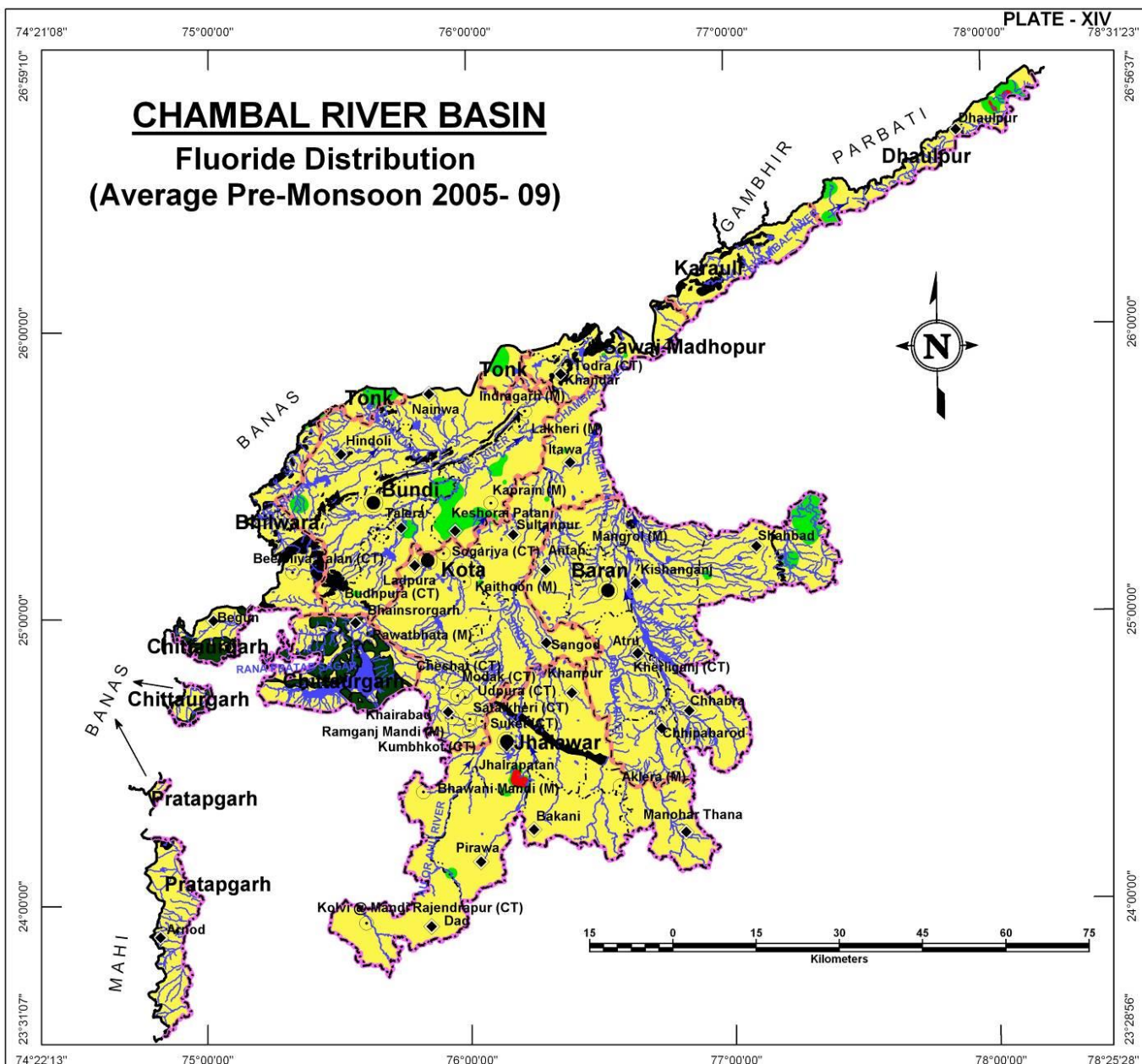
The Fluoride concentration map (Plate – XIV) displays two isolated patches of high fluoride concentration (>3 mg/l) which are located around Jhalrapatan in the southern part and north of Dhaulpur in the extreme northeastern part of the basin. Apart from these two patches which together constitute less than 2% of the basin and the moderate concentration areas (1.5 to 3.0 mg/l) that occupy around 3.5% of the basin area, rest of the basin falls under low concentration zone (< 1.5 mg/l). Therefore, by and large the basin seems to be not much affected by Fluoride problem.

| Fluoride Ranges<br>(mg/l)<br>(Ave. of years 2005-09) | District wise area coverage (sq km) |       |          |       |         |       |              |       |          |       |          |       |         |       |         |       |            |       |                |       |       |       | Total Area<br>(sq km) |
|------------------------------------------------------|-------------------------------------|-------|----------|-------|---------|-------|--------------|-------|----------|-------|----------|-------|---------|-------|---------|-------|------------|-------|----------------|-------|-------|-------|-----------------------|
|                                                      | Baran                               |       | Bhilwara |       | Bundi   |       | Chittaurgarh |       | Dhaulpur |       | Jhalawar |       | Karauli |       | Kota    |       | Pratapgarh |       | Sawai Madhopur |       | Tonk  |       |                       |
|                                                      | Area                                | % age | Area     | % age | Area    | % age | Area         | % age | Area     | % age | Area     | % age | Area    | % age | Area    | % age | Area       | % age | Area           | % age | Area  | % age |                       |
| < 1.5                                                | 6,725.7                             | 96.5  | 811.2    | 98.9  | 4,996.6 | 92.1  | 1,327.7      | 100.0 | 672.1    | 79.5  | 6,122.2  | 98.7  | 710.2   | 100.0 | 5,105.6 | 99.7  | 874.6      | 100.0 | 829.4          | 99.4  | 306.7 | 68.8  | 28,482.0              |
| 1.5-3.0                                              | 240.4                               | 3.5   | 9.4      | 1.1   | 428.8   | 7.9   | -            | -     | 155.0    | 18.4  | 49.8     | 0.8   | 0.1     | -     | 16.7    | 0.3   | -          | -     | 5.2            | 0.6   | 139.3 | 31.2  | 1,044.7               |
| > 3.0                                                | -                                   | -     | -        | -     | -       | -     | -            | -     | 17.8     | 2.1   | 33.3     | 0.5   | -       | -     | -       | -     | -          | -     | -              | -     | -     | -     | 51.1                  |
| Total                                                | 6,966.1                             | 100.0 | 820.6    | 100.0 | 5,425.4 | 100.0 | 1,327.7      | 100.0 | 844.9    | 100.0 | 6,205.3  | 100.0 | 710.3   | 100.0 | 5,122.3 | 100.0 | 874.6      | 100.0 | 834.6          | 100.0 | 446.0 | 100.0 | 29,577.8              |

## NITRATE DISTRIBUTION

High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate – XV shows distribution of Nitrate in groundwater. Except for the central part of the basin i.e., within Baran – Kota – Bundi – Tonk region, that has low concentration of Nitrates in ground water, the basin has scattered patches of high (> 100 mg/l) and moderately high (50-100 mg/l) concentration of Nitrates in ground water. Such high concentration patches are seen in east, south and northwestern parts of the basin around Baran, Jhalawar, Mandi Rajdrapur, Bhilwara, Hindoli, Khandar, Dhaulpur etc.

| Nitrate Ranges<br>(mg/l)<br>(Ave. of years 2005-09) | District wise area coverage (sq km) |       |          |       |         |       |              |       |          |       |          |       |         |       |         |       |            |       |                |       |       |       | Total Area<br>(sq km) |
|-----------------------------------------------------|-------------------------------------|-------|----------|-------|---------|-------|--------------|-------|----------|-------|----------|-------|---------|-------|---------|-------|------------|-------|----------------|-------|-------|-------|-----------------------|
|                                                     | Baran                               |       | Bhilwara |       | Bundi   |       | Chittaurgarh |       | Dhaulpur |       | Jhalawar |       | Karauli |       | Kota    |       | Pratapgarh |       | Sawai Madhopur |       | Tonk  |       |                       |
|                                                     | Area                                | % age | Area     | % age | Area    | % age | Area         | % age | Area     | % age | Area     | % age | Area    | % age | Area    | % age | Area       | % age | Area           | % age | Area  | % age |                       |
| < 50                                                | 5,374.3                             | 77.1  | 500.7    | 61.0  | 4,355.8 | 80.3  | 842.5        | 63.5  | 721.2    | 85.4  | 3,060.7  | 49.3  | 197.8   | 27.8  | 3,980.3 | 77.7  | 324.6      | 37.1  | 543.9          | 65.2  | 239.8 | 53.8  | 20,141.6              |
| 50-100                                              | 1,358.3                             | 19.5  | 264.7    | 32.3  | 932.5   | 17.2  | 485.2        | 36.5  | 104.2    | 12.3  | 2,822.7  | 45.5  | 512.5   | 72.2  | 941.7   | 18.4  | 546.0      | 62.4  | 247.9          | 29.7  | 194.6 | 43.6  | 8,410.3               |
| > 100                                               | 233.5                               | 3.4   | 55.2     | 6.7   | 137.1   | 2.5   | -            | -     | 19.5     | 2.3   | 321.9    | 5.2   | -       | -     | 200.3   | 3.9   | 4.0        | 0.5   | 42.8           | 5.1   | 11.6  | 2.6   | 1,025.9               |
| Total                                               | 6,966.1                             | 100.0 | 820.6    | 100.0 | 5,425.4 | 100.0 | 1,327.7      | 100.0 | 844.9    | 100.0 | 6,205.3  | 100.0 | 710.3   | 100.0 | 5,122.3 | 100.0 | 874.6      | 100.0 | 834.6          | 100.0 | 446.0 | 100.0 | 29,577.8              |





## DEPTH TO BEDROCK

## CHAMBAL RIVER BASIN

Plate – XVI represents depth to bed rock distribution below ground level. A perusal of map reveals that in general, the depth to bedrock in the basin is very shallow being encountered at a depth of less than 40m bgl indicating limited thickness of weathered/fractured material or alluvial cover present in the basin in order to form promising aquifers. The bedrock is deepest in the northeastern edge of the basin around Dhaulpur where depths of >60m bgl are seen reaching up to even 100m bgl.

| Depth to Bedrock<br>(m bgl) | District wise area coverage (sq km) |              |                |                |              |                |              |                |              |                   |              | Total Area<br>(sq km) |
|-----------------------------|-------------------------------------|--------------|----------------|----------------|--------------|----------------|--------------|----------------|--------------|-------------------|--------------|-----------------------|
|                             | Baran                               | Bhilwara     | Bundi          | Chittaurgarh   | Dhaulpur     | Jhalawar       | Karauli      | Kota           | Pratapgarh   | Sawai<br>Madhopur | Tonk         |                       |
| < 20                        | 4,792.1                             | 396.9        | 3,293.5        | 871.7          | -            | 4,352.2        | 70.3         | 3,290.1        | 106.2        | -                 | 188.3        | 17,361.3              |
| 20-40                       | 2,145.5                             | 350.9        | 1,968.5        | 259.5          | 3.7          | 1,756.1        | 474.4        | 1,832.2        | 768.4        | 802.9             | 257.3        | 10,619.4              |
| 40-60                       | 28.5                                | 57.2         | 163.4          | 196.5          | 394.5        | 97.0           | 165.6        | -              | -            | 31.7              | 0.4          | 1,134.8               |
| 60-80                       | -                                   | 15.6         | -              | -              | 273.2        | -              | -            | -              | -            | -                 | -            | 288.8                 |
| 80-100                      | -                                   | -            | -              | -              | 102.0        | -              | -            | -              | -            | -                 | -            | 102.0                 |
| > 100                       | -                                   | -            | -              | -              | 71.5         | -              | -            | -              | -            | -                 | -            | 71.5                  |
| <b>Total</b>                | <b>6,966.1</b>                      | <b>820.6</b> | <b>5,425.4</b> | <b>1,327.7</b> | <b>844.9</b> | <b>6,205.3</b> | <b>710.3</b> | <b>5,122.3</b> | <b>874.6</b> | <b>834.6</b>      | <b>446.0</b> | <b>29,577.8</b>       |

## UNCONFINED AQUIFER

Hydrogeological properties are different for alluvial and hard rock aquifers and therefore, this aquifer has been mapped as two separate regions viz, unconfined aquifers in alluvial and in hard rock areas. Good aquifer thickness is seen in Bundi, Kota and Dhaulpur districts whereas the rest of the districts largely do not have any significant alluvial aquifers. In hardrock areas however, the weathered/fractured zone is widespread and forms aquifers in the basin as seen by good thicknesses of 20-70m is seen in Bhilwara, Bundi, Dhaulpur, Jhalawar and Karauli.

### Alluvial areas

| District Name  | District wise area (sq km) within<br>Aquifer thickness range (m) |              | Total Area<br>(sq km) |
|----------------|------------------------------------------------------------------|--------------|-----------------------|
|                | < 10                                                             | 10-20        |                       |
| Baran          | 326.9                                                            | -            | 326.9                 |
| Bhilwara       | -                                                                | -            | -                     |
| Bundi          | 747.4                                                            | 516.4        | 1,263.8               |
| Chittaurgarh   | -                                                                | -            | -                     |
| Dhaulpur       | 446.6                                                            | 102.7        | 549.3                 |
| Jhalawar       | -                                                                | -            | -                     |
| Karauli        | 142.8                                                            | -            | 142.8                 |
| Kota           | 1,342.4                                                          | 206.9        | 1,549.3               |
| Pratapgarh     | -                                                                | -            | -                     |
| Sawai Madhopur | 192.7                                                            | 14.9         | 207.6                 |
| Tonk           | -                                                                | 20.3         | 20.3                  |
| <b>Total</b>   | <b>3,198.8</b>                                                   | <b>861.2</b> | <b>4,060.0</b>        |

### Hardrock areas:

| District Name     | District wise area (sq km) within Aquifer thickness range (m) |                |                |              |              |              |             |             |             |          | Total Area<br>(sq km) |
|-------------------|---------------------------------------------------------------|----------------|----------------|--------------|--------------|--------------|-------------|-------------|-------------|----------|-----------------------|
|                   | <10                                                           | 10-20          | 20-30          | 30-40        | 40-50        | 50-60        | 60-70       | 70-80       | 80-90       | > 90     |                       |
| Baran             | 6,498.9                                                       | 140.3          | -              | -            | -            | -            | -           | -           | -           | -        | 6,639.2               |
| Bhilwara          | 277.4                                                         | 245.4          | 123.6          | 78.4         | 57.6         | 33.9         | 4.3         | -           | -           | -        | 820.6                 |
| Bundi             | 3,039.1                                                       | 777.6          | 241.0          | 88.5         | 15.4         | -            | -           | -           | -           | -        | 4,161.6               |
| Chittaur-garh     | 1,092.8                                                       | 234.0          | 0.9            | -            | -            | -            | -           | -           | -           | -        | 1,327.7               |
| Dhaulpur          | -                                                             | -              | -              | 44.8         | 134.9        | 48.5         | 29.5        | 23.4        | 14.5        | -        | 295.6                 |
| Jhalawar          | 4,031.3                                                       | 1,126.7        | 683.0          | 245.4        | 72.5         | 44.1         | 2.3         | -           | -           | -        | 6,205.3               |
| Karauli           | -                                                             | 451.8          | 68.8           | 13.2         | 24.8         | 8.9          | -           | -           | -           | -        | 567.5                 |
| Kota              | 3,570.8                                                       | 2.2            | -              | -            | -            | -            | -           | -           | -           | -        | 3,573.0               |
| Pratapgarh        | 718.2                                                         | 146.6          | 9.8            | -            | -            | -            | -           | -           | -           | -        | 874.6                 |
| Sawai<br>Madhopur | 506.4                                                         | 110.6          | 10.0           | -            | -            | -            | -           | -           | -           | -        | 627.0                 |
| Tonk              | 277.3                                                         | 121.0          | 18.5           | 7.7          | 1.2          | -            | -           | -           | -           | -        | 425.7                 |
| <b>Total</b>      | <b>20,012.2</b>                                               | <b>3,356.2</b> | <b>1,155.6</b> | <b>478.0</b> | <b>306.4</b> | <b>135.4</b> | <b>36.1</b> | <b>23.4</b> | <b>14.5</b> | <b>-</b> | <b>25,517.8</b>       |



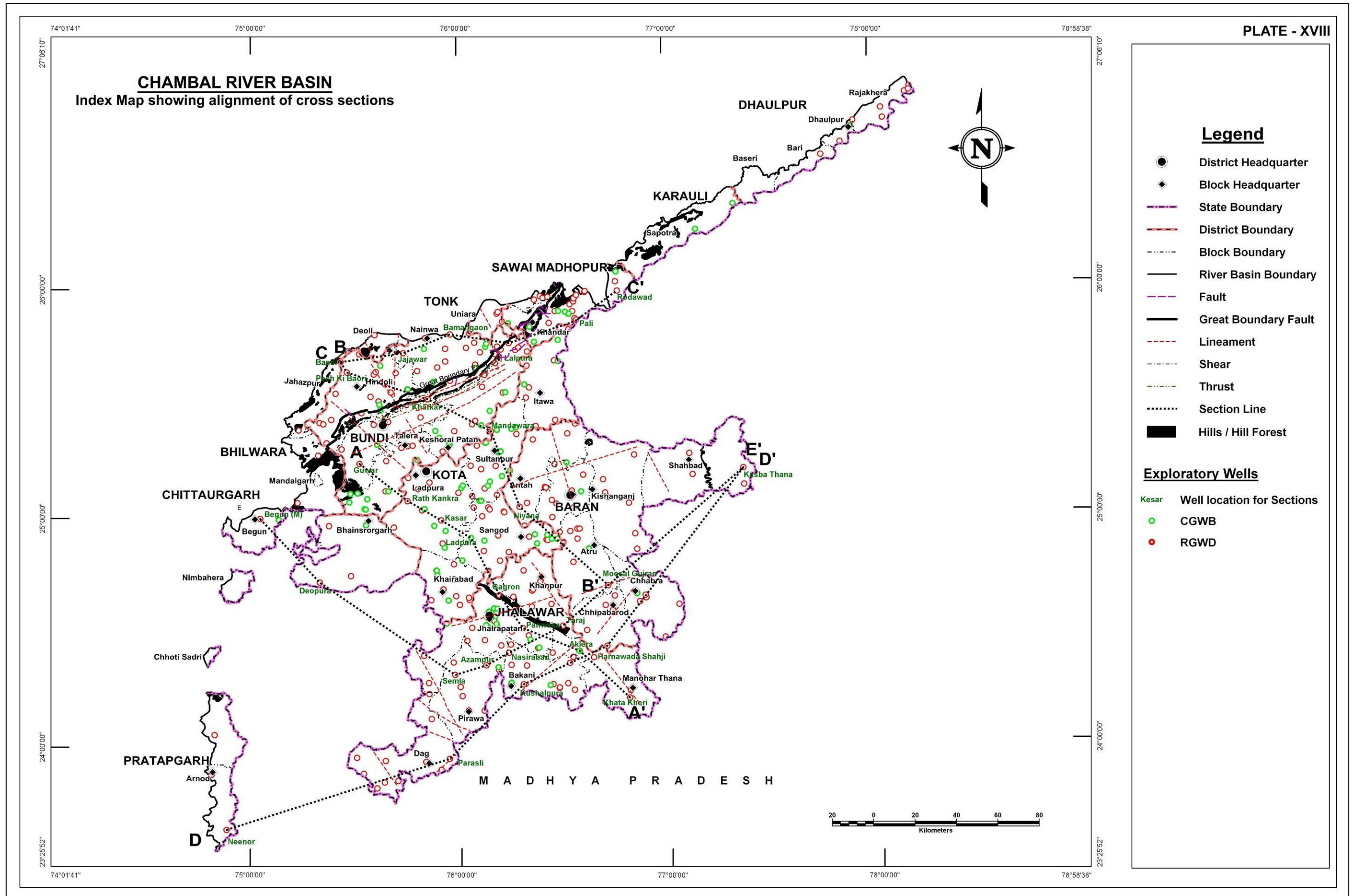


## CROSS SECTIONS

## CHAMBAL RIVER BASIN

Several hydrogeologic cross sections have been drawn to better depict the sub-surface distribution of lithology. These sections have been overlaid with geological maps and structural faults if there are any have been transferred for verification of their impact on sub-surface material disposition. The alignment of the cross sections is shown in Plate – XVIII and corresponding sections are presented in Plates – XIX to XXIII. The broad alignment of the sections is as given below:

| Name of Section Line | Orientation  |
|----------------------|--------------|
| Section AA'          | NW – SE      |
| Section BB'          | NW – SE      |
| Section CC'          | SW – NE      |
| Section DD'          | NW – SE      |
| Section EE'          | NW – SE – NE |





## CROSS SECTIONS

## CHAMBAL RIVER BASIN

### Section A-A':

The A-A' section (Plate – XIX) trends in NW-SE direction stretching to a distance of about 175 kms. The section depicts the disposition of different layers of sand, clay, limestone along with basalt. The lithologs of 8 boreholes along with surrounding well information is taken while preparing the section. On perusal of the cross section, it is revealed that while the southeastern part of the basin subsurface is predominantly Basaltic in nature, central part is dominated by Shales and west by Sandstones of Vindhyan period.

The water level varies from 290m amsl to 360m amsl following the surface topography as observed from the 2010 pre monsoon season.

### Section B-B':

The B-B' (Plate – XX) section is has been chosen to represent a NW-SE profile and covers a length of about 225km, perpendicular to the Chambal river flow direction. The lithologs of 6 boreholes along with surrounding well information is taken while preparing the section. In the central part of the section where Chambal river is intersected, about 30m of alluvial material is deposited. At 50 km of the profile a fault i.e., Great Boundary Fault (F) is present that separates phyllites to its west from Vindhyan shales and sandstones to its east.

Ground water level varies from 225m amsl to 325m amsl in this profile.

PLATE - XIX

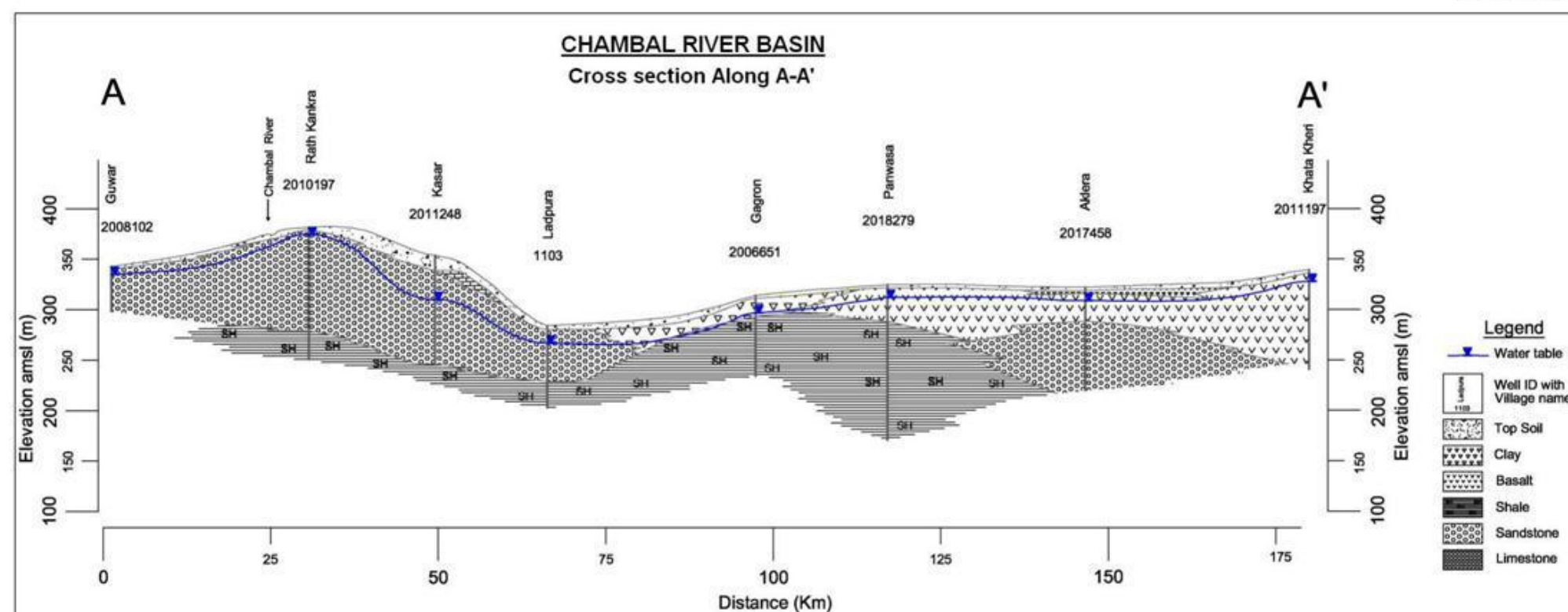
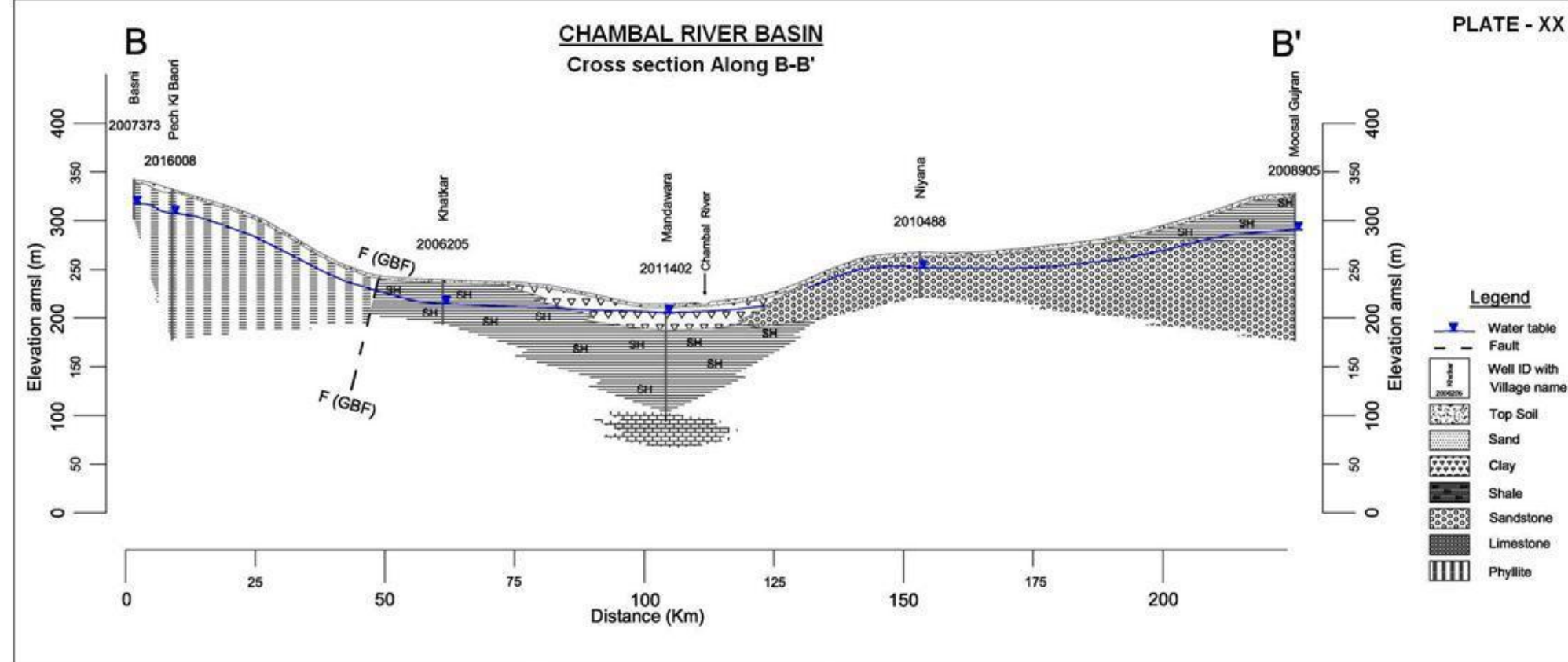


PLATE - XX





## CROSS SECTIONS

## CHAMBAL RIVER BASIN

### Section C-C':

The C-C' (Plate – XXI) section has been selected across the basin trending SW-NE, for a distance of about 150 km in the western part of the basin. The lithologs of 6 boreholes along with surrounding well information is taken while preparing the section. Roughly in the middle of the section a fault is seen which actually is the Great Boundary Fault (F) and cuts across the section. To the west of this fault, phyllites are present in the subsurface whereas in the east of the fault, Vindhyan shales and sandstones are present. Within the Vindhyan, the Chambal river flows and alluvial clay is deposited alongside the river to further east.

The ground water level in this section varies from 175m amsl to 320m amsl.

### Section D-D':

The section D-D' (Plate – XXII) is the longest of the sections plotted in the area and cover a length of 325 km trending in SW-NE direction. The lithologs of 5 boreholes along with surrounding well information is taken into consideration while preparing the section. Basalt is predominant in the 0-200 km length of the section while 200 to 325 km stretch depicts mainly the alternating layers of shale and sandstone of Vindhyan period in the sub-surface.

Ground water level varies from 325m amsl to 500m amsl in this section.

PLATE - XXI

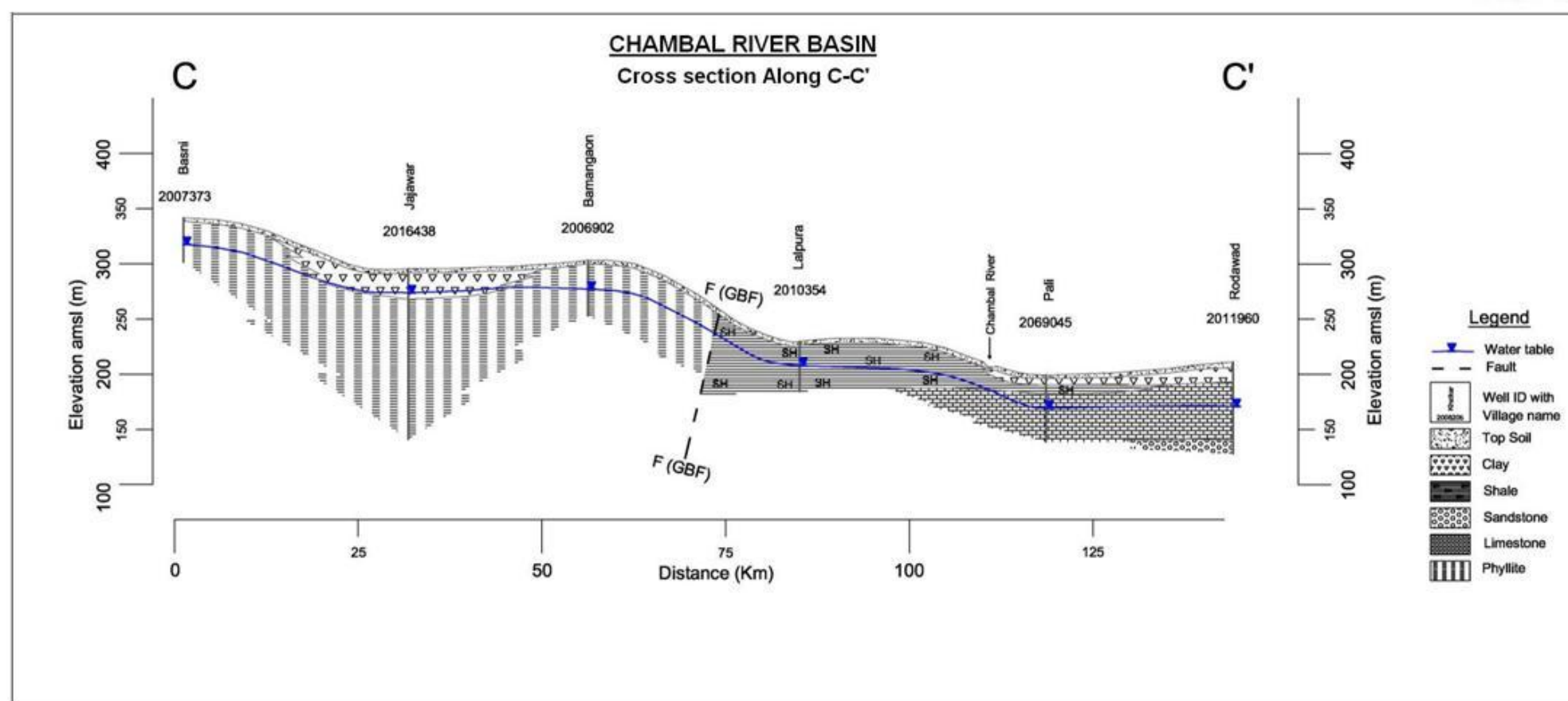
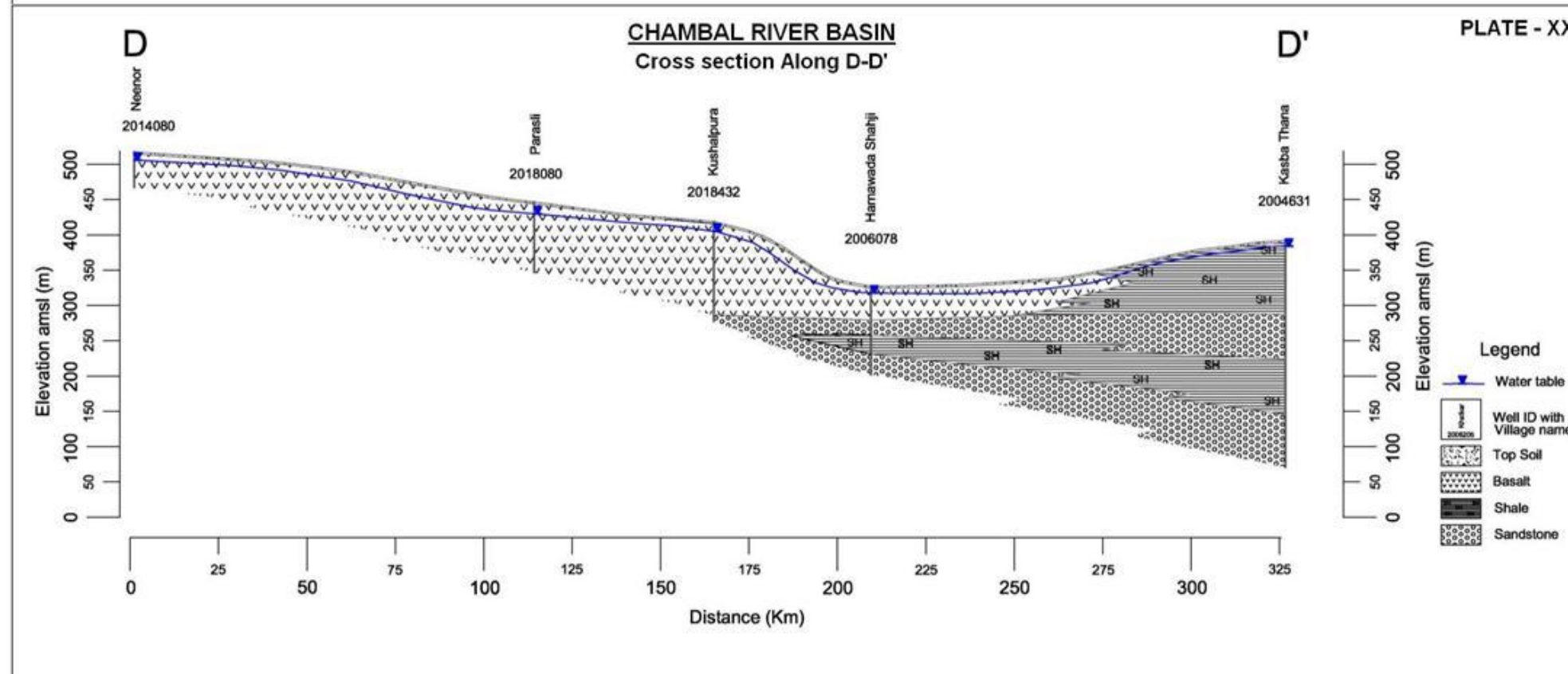


PLATE - XXII





## CROSS SECTIONS

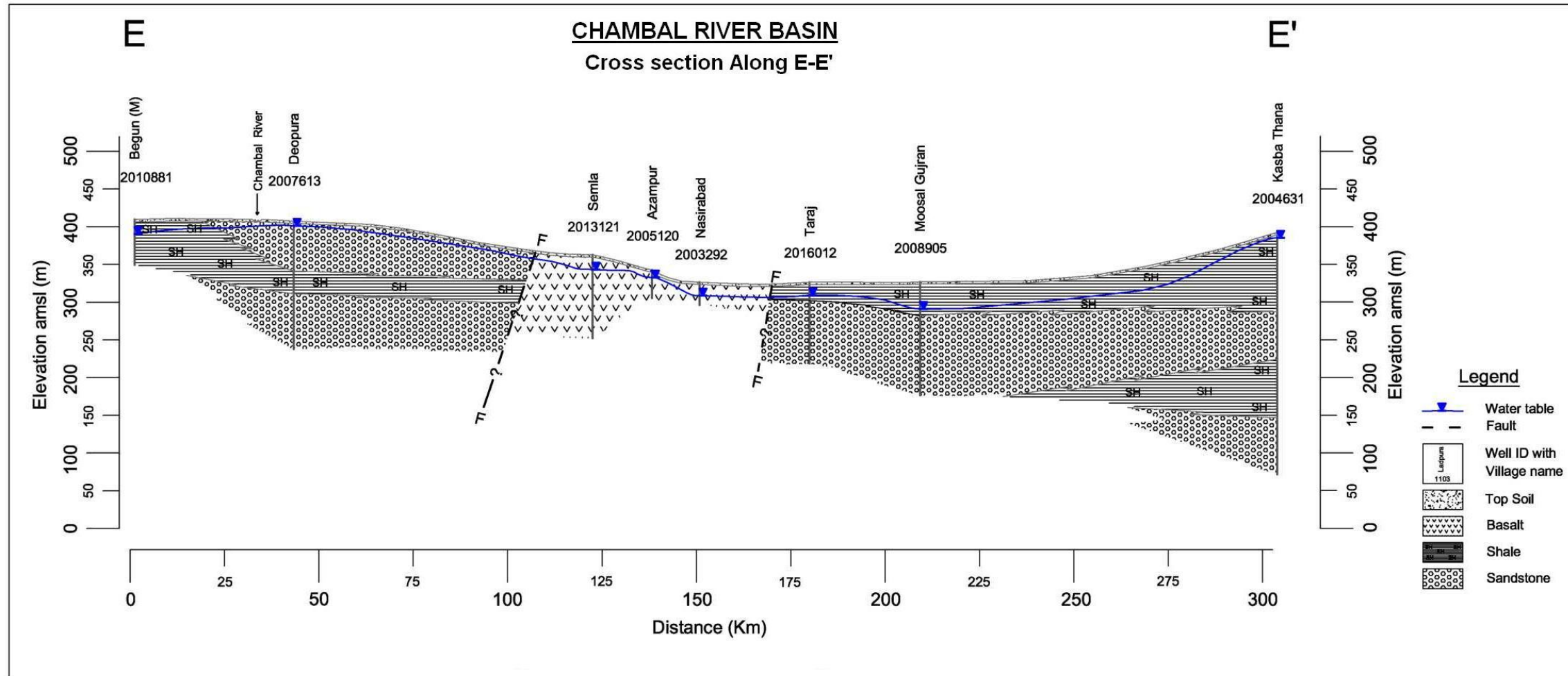
## CHAMBAL RIVER BASIN

### Section E-E’:

The E-E’ section (Plate – XXIII) is the second largest section plotted in the area, covering a length of 300 km that initially trends in NW-SE and later turns NE. The lithologs of 8 boreholes along with surrounding well information is taken while preparing the section. Vindhyan Super Group rocks of Shale intercalated with sandstone are present in the sub-surface overlain by thin cover of top soil. Two tentative faults apparently separate the central Basaltic intrusive deposits with surrounding Vindhyan.

Ground water level in this section varies from 300 m amsl to 400 m amsl.

PLATE - XXIII



## 3D MODEL OF AQUIFERS

## CHAMBAL RIVER BASIN

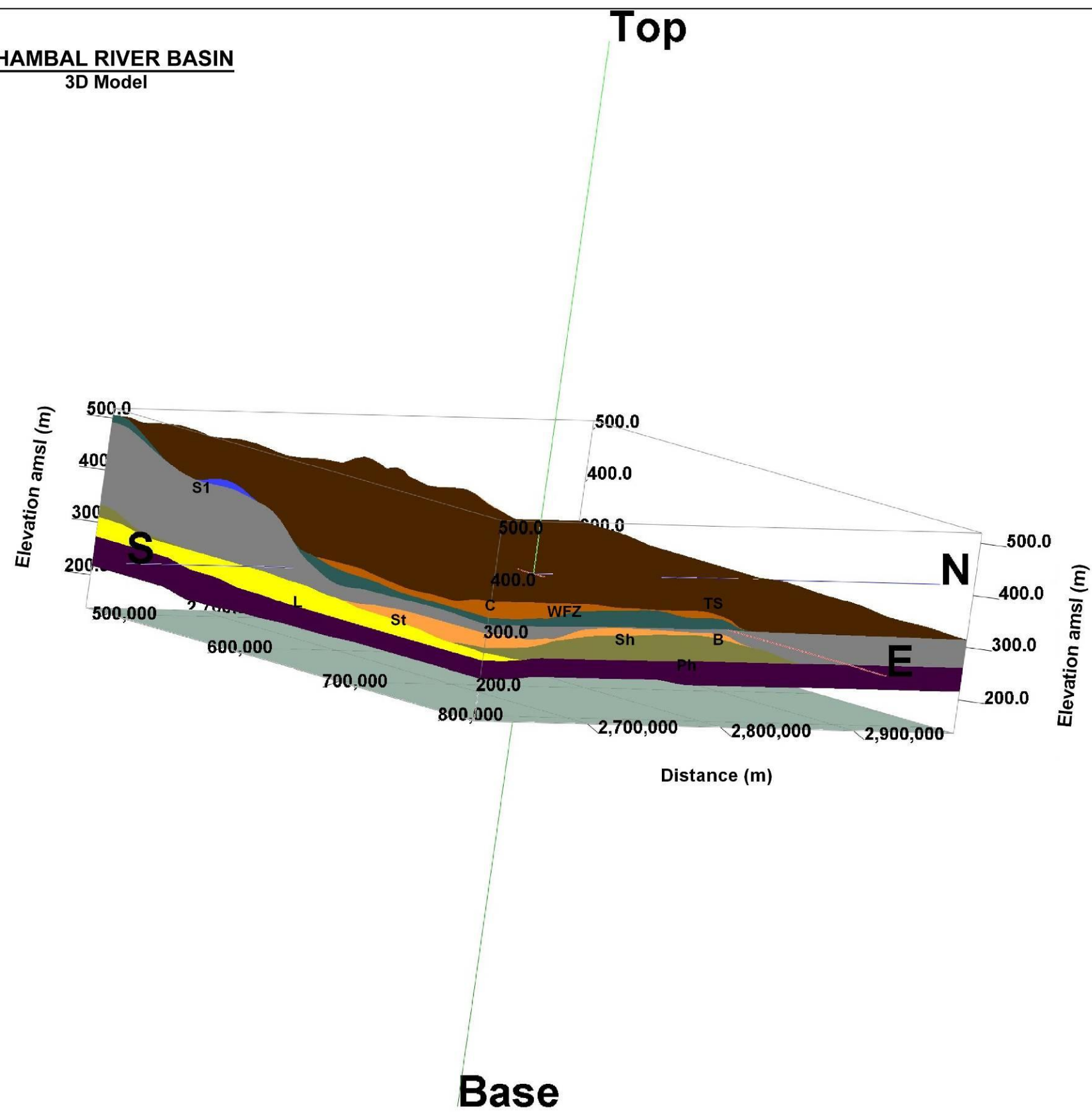
The continuous litho-stratigraphic model has been developed using Rockworks software for the Chambal river basin using interpolation techniques on the data of scattered wells given as an input.

The (Plate – XXIV) presents the 3D model depicting the various litho-stratigraphic units in the entire river basin. With this model it is apparent that beneath the top soil there is a persistent weathered and fractured hard rock material is present that acts as unconfined aquifer. The limestone aquifer is found overlain by basalt in the southern part of the basin. The sandstone formation is present in the south eastern part of the basin is overlain by basalt and underlain by shale formation. Phyllite is present all over the basin as a continuous formation is overlain by limestone in the south and shale in the east and basalt in rest of the area.



PLATE - XXIV

**CHAMBAL RIVER BASIN**  
3D Model



**Legend**

|     |                            |
|-----|----------------------------|
| TS  | Top Soil                   |
| S1  | Sand                       |
| C   | Clay                       |
| S2  | Sand                       |
| WFZ | Weathered & Fractured Zone |
| B   | Basalt                     |
| St  | Sandstone                  |
| Sh  | Shale                      |
| L   | Limestone                  |
| Ph  | Phyllite                   |

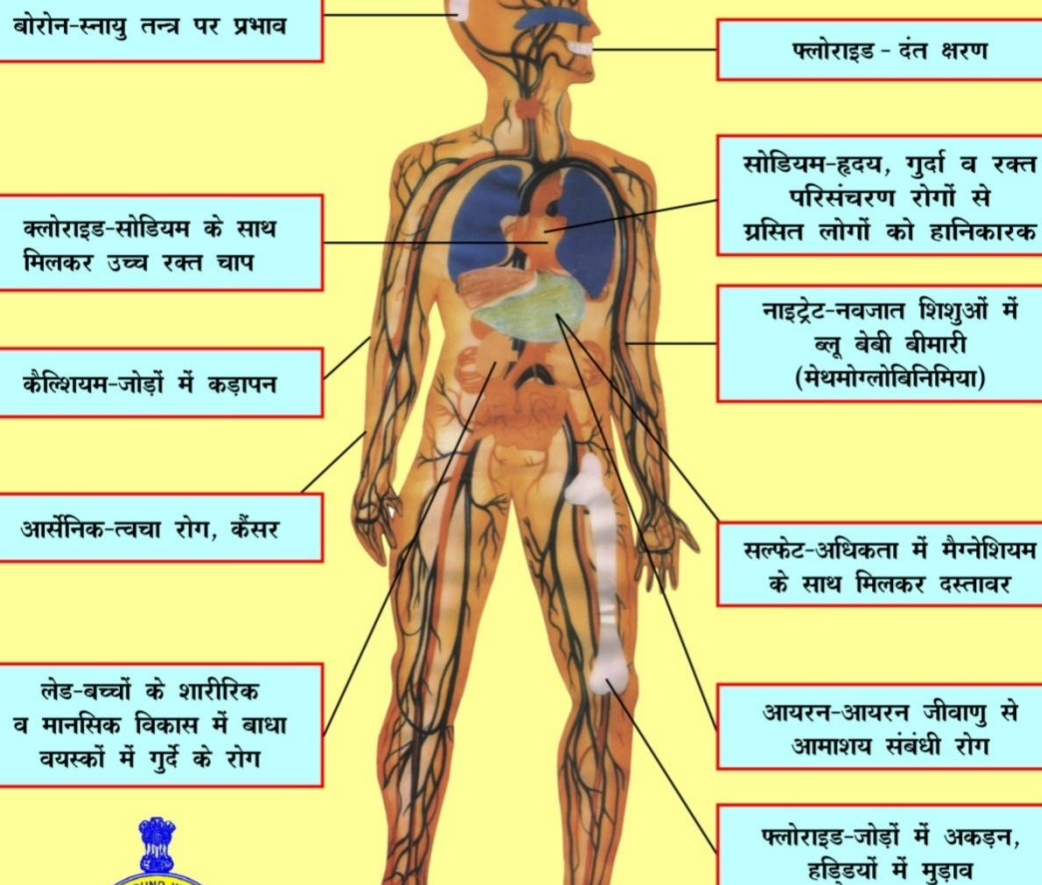
## Glossary of terms

| S. No. | Technical Terms        | Definition                                                                                                                                   |
|--------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | AQUIFER                | A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.          |
| 2      | ARID CLIMATE           | Climate characterized by high evaporation and low precipitation.                                                                             |
| 3      | ARTIFICIAL RECHARGE    | Addition of water to a groundwater reservoir by man-made activity                                                                            |
| 4      | CLIMATE                | The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region. |
| 5      | CONFINED AQUIFER       | A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.                 |
| 6      | CONTAMINATION          | Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.                      |
| 7      | DRAWDOWN               | The drawdown is the depth by which water level is lowered.                                                                                   |
| 8      | FRESH WATER            | Water suitable for drinking purpose.                                                                                                         |
| 9      | GROUND WATER           | Water found below the land surface.                                                                                                          |
| 10     | GROUND WATER BASIN     | A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.                                           |
| 11     | GROUNDWATER RECHARGE   | The natural infiltration of surface water into the ground.                                                                                   |
| 12     | HARD WATER             | The water which does not produce sufficient foam with soap.                                                                                  |
| 13     | HYDRAULIC CONDUCTIVITY | A constant that serves as a measure of permeability of porous medium.                                                                        |
| 14     | HYDROGEOLOGY           | The science related with the ground water.                                                                                                   |
| 15     | HUMID CLIMATE          | The area having high moisture content.                                                                                                       |
| 16     | ISOHYET                | A line of equal amount of rainfall.                                                                                                          |
| 17     | METEOROLOGY            | Science of the atmosphere.                                                                                                                   |
| 18     | PERCOLATION            | It is flow through a porous substance.                                                                                                       |
| 19     | PERMEABILITY           | The property or capacity of a soil or rock for transmitting water.                                                                           |
| 20     | pH                     | Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).                               |
| 21     | PIEZOMETRIC HEAD       | Elevation to which water will rise in a piezometers.                                                                                         |
| 22     | RECHARGE               | It is a natural or artificial process by which water is added from outside to the aquifer.                                                   |
| 23     | SAFE YIELD             | Amount of water which can be extracted from groundwater without producing undesirable effect.                                                |
| 24     | SALINITY               | Concentration of dissolved salts.                                                                                                            |
| 25     | SEMI-ARID              | An area is considered semiarid having annual rainfall between 10-20 inches.                                                                  |
| 26     | SEMI-CONFINED AQUIFER  | Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.                                                                  |
| 27     | SPECIFIC YIELD         | Quantity of water which is released by a formation after its complete saturation.                                                            |
| 28     | TOTAL DISSOLVED SOLIDS | Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.                                  |

(Contd...)

| S. No. | Technical Terms            | Definition                                                                                                                                                                                                                   |
|--------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29     | TRANSMISSIBILITY           | It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability. |
| 30     | UNCONFINED AQUIFER         | A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.                                                                                                              |
| 31     | UNSATURATED ZONE           | The zone below the land surface in which pore space contains both water and air.                                                                                                                                             |
| 32     | WATER CONSERVATION         | Optimal use and proper storage of water.                                                                                                                                                                                     |
| 33     | WATER RESOURCES            | Availability of surface and ground water.                                                                                                                                                                                    |
| 34     | WATER RESOURCES MANAGEMENT | Planned development, distribution and use of water resources.                                                                                                                                                                |
| 35     | WATER TABLE                | Water table is the upper surface of the zone of saturation at atmospheric pressure.                                                                                                                                          |
| 36     | ZONE OF SATURATION         | The ground in which all pores are completely filled with water.                                                                                                                                                              |
| 37     | ELECTRICAL CONDUCTIVITY    | Flow of free ions in the water at 25C mu/cm.                                                                                                                                                                                 |
| 38     | CROSS SECTION              | A Vertical Projection showing sub-surface formations encountered in a specific plane.                                                                                                                                        |
| 39     | 3-D PICTURE                | A structure showing all three dimensions i.e. length, width and depth.                                                                                                                                                       |
| 40     | GWD                        | Ground Water Department                                                                                                                                                                                                      |
| 41     | CGWB                       | Central Ground Water Board                                                                                                                                                                                                   |
| 42     | CGWA                       | Central Ground Water Authority                                                                                                                                                                                               |
| 43     | SWRPD                      | State Water Resources Planning Department                                                                                                                                                                                    |
| 44     | EU-SPP                     | European Union State Partnership Programme                                                                                                                                                                                   |
| 45     | TOPOGRAPHY                 | Details of drainage lines and physical features of land surface on a map.                                                                                                                                                    |
| 46     | GEOLOGY                    | The science related with the Earth.                                                                                                                                                                                          |
| 47     | GEOMORPHOLOGY              | The description and interpretation of land forms.                                                                                                                                                                            |
| 48     | PRE MONSOON SURVEY         | Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)                                                                                                 |
| 49     | POST-MONSOON SURVEY        | Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)                                                                                          |
| 50     | PIEZOMETER                 | A non-pumping small diameter bore hole used for monitoring of static water level.                                                                                                                                            |
| 51     | GROUND WATER FLUCTUATION   | Change in static water level below ground level.                                                                                                                                                                             |
| 52     | WATER TABLE                | The static water level found in unconfined aquifer.                                                                                                                                                                          |
| 53     | DEPTH OF BED ROCK          | Hard & compact rock encountered below land Surface.                                                                                                                                                                          |
| 54     | G.W. MONITORING STATION    | Dug wells selected on grid basis for monitoring of state water level.                                                                                                                                                        |
| 55     | EOLIAN DEPOSITS            | Wind-blown sand deposits                                                                                                                                                                                                     |





**भूजल अमूल्य है इसे प्रदूषित न करें।**

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## Myths and Facts about Ground Water

| S No | Myths                                                                                                                                                                        | Facts                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | What is Ground Water <ul style="list-style-type: none"> <li>• an underground lake</li> <li>• a net work of underground rivers</li> <li>• a bowl filled with water</li> </ul> | Water which occurs below the land in geological formations/rocks is Ground water                                                                                                       |
| 2    | Ground Water occurs everywhere beneath the Land Surface                                                                                                                      | Not really, it depends on the nature of rock formation                                                                                                                                 |
| 3    | There is a relationship between ground water and surface water                                                                                                               | Not all the places. Near streams/rivers there is relation                                                                                                                              |
| 4    | Groundwater is not renewable resource                                                                                                                                        | It is renewable source and every year it is being recharged through rain/applied irrigation etc                                                                                        |
| 5    | Ground water is unlimited and deeper you drill more discharge                                                                                                                | It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper                                                                         |
| 6    | Ground Water moves rapidly                                                                                                                                                   | The movement of ground water is very slow                                                                                                                                              |
| 7    | Ground water pumped from wells is thousands of years old                                                                                                                     | Generally the ground water being tapped through wells is a few years old                                                                                                               |
| 8    | If water taste good—it is safe to drink                                                                                                                                      | It may have other chemicals e.g. fluoride, nitrates etc which are harmful                                                                                                              |
| 9    | Water from free flowing tube wells is very pure                                                                                                                              | This water can also be contaminated so test before use                                                                                                                                 |
| 10   | If I recharge my TW/DW/HP it will not benefit me                                                                                                                             | It will also benefit you and also adjoining wells                                                                                                                                      |
| 11   | There is no static ground water resources in Rajasthan                                                                                                                       | Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge                                                   |
| 12   | I cannot meet annual cooking and drinking water requirement by rain water harvesting                                                                                         | The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement |
| 13   | You can increase ground water recharge                                                                                                                                       | This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures                                                        |
| 14   | You cannot use abandoned TW/HP/DW for ground water recharge                                                                                                                  | These should be used as recharge structures as harvested rain water is directly put into GW reservoir                                                                                  |
| 15   | Putting waste near HP/TW will not cause any problem                                                                                                                          | Such actions will pollute wells and water                                                                                                                                              |



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